je vais t'envoyer une simulation et aide moi à la faire etape par étape alors c une simulation ros2 gazebo, premierement il y a tugbot_description, tugbot_gazebo,tugbot_navigation2, tugbot_ros2_pkgs, tugbot_slam dans tugbot_description il y a launch dans lequel il y a tugbot_description.launch.py dans lequel il y a ecris from launch import LaunchDescription

from launch_ros.actions import Node

from launch.substitutions import LaunchConfiguration

from launch.actions import DeclareLaunchArgument

from ament_index_python.packages import get_package_share_directory

import os

def generate_launch_description():

urdf_path = os.path.join(get_package_share_directory('tugbot_description'), 'urdf', 'model.urdf')

return LaunchDescription([

DeclareLaunchArgument(

'urdf_file',

default_value=urdf_path,

description='Full path to the URDF file to load'

),

Node(

package='robot_state_publisher',

executable='robot_state_publisher',

arguments=[LaunchConfiguration('urdf_file')],

output='screen'

)

]) il y aaussi meshes (dans lequel il y a base, gripper2 , light_link, wheel, VLP16_base_1.dae et VLP16_base_2.dae et VLP16_scan.dae,) il ya aussi urdf (model.sdf (<?xml version='1.0' encoding='ASCII'?>

<sdf version="1.6">

<model name="tugbot">

<self_collide>false</self_collide>

<pose>0 0 0.2 0 0 0</pose>

<static>false</static>

<!-- BASE -->

```
<link name="base_link">
<pose>0 0 0 0 0 0</pose>
<inertial>
<pose>-0.1 0 0.1 0 0 0</pose>
<mass>46.2</mass>
<inertia>
<ixx>1.55925</ixx>
<ixy>0</ixy>
<ixz>0</ixz>
<iyy>1.55925</iyy>
<iyz>0</iyz>
<izz>1.55925</izz>
</inertia>
</inertial>
<collision name="base_link_collision">
<pose>0 0 0 0 0 0</pose>
<geometry>
<mesh>
<uri>meshes/base/tugbot_simp.stl</uri>
</mesh>
</geometry>
</collision>
<visual name="base_link_visual">
<pose>0 0 0 0 0 0</pose>
<geometry>
<mesh>
<uri>meshes/base/tugbot_simp.dae</uri>
</mesh>
</geometry>
```

```xml
</visual>

<visual name="movai_logo_visual">

<pose>0 0 0 0 0 0</pose>

<geometry>

<mesh>

<uri>meshes/base/movai_logo.dae</uri>

</mesh>

</geometry>

<material>

<diffuse>1.0 1.0 1.0</diffuse>

<pbr>

<metal>

<albedo_map>meshes/base/movai-logo.png</albedo_map>

</metal>

</pbr>

</material>

</visual>

<visual name="light_link_visual">

<pose>-0.1 0 0.1945 0 1.57079632679 0</pose>

<geometry>

<mesh>

<scale>1.0 1.0 1.0</scale>

<uri>meshes/light_link/light_led.stl</uri>

</mesh>

</geometry>

<material>

<ambient>0.0 0.0 0 1</ambient>

<diffuse>0.0 0.0 0 1</diffuse>

<specular>0 0 0 1</specular>
```

```
<emissive>0 1 0 1</emissive>
</material>
</visual>
<visual name="warnign_light_visual">
<pose>-0.099 0 0.195 0 1.57079632679 0</pose>
<geometry>
<mesh>
<scale>1.02 1.02 1.02</scale>
<uri>meshes/light_link/light.stl</uri>
</mesh>
</geometry>
<material>
<ambient>0.1 0.05 0 0.5</ambient>
<diffuse>1 0.5 0 0.5</diffuse>
<specular>1 0.5 0 0.5</specular>
<emissive>1 0.5 0 0.5</emissive>
</material>
</visual>
</link>
<link name="imu_link">
<pose>0.14 0.02 0.25 0 0 -1.57</pose>
<sensor name="imu" type="imu">
<update_rate>30</update_rate>
<visualize>false</visualize>
</sensor>
</link>
<joint name="imu_link_joint" type="fixed">
<child>imu_link</child>
<parent>base_link</parent>
```

```
</joint>
<link name="warnign_light">
<pose>-0.185 0 0.46 0 0 0</pose>
<inertial>
<pose>0 0 0 0 0 0</pose>
<mass>0.1</mass>
<inertia>
<ixx>0.000016875</ixx>
<ixy>0</ixy>
<ixz>0</ixz>
<iyy>0.00000001</iyy>
<iyz>0</iyz>
<izz>0.000016875</izz>
</inertia>
</inertial>
<light name="warnign_light_lamp" type="spot">
<pose>-0.04 0 0 0 1.57079632679 0</pose>
<attenuation>
<range>5</range>
<linear>0</linear>
<constant>0.1</constant>
<quadratic>0.0025</quadratic>
</attenuation>
<diffuse>1 0.5 0 1</diffuse>
<specular>1 0.5 0 1</specular>
<spot>
<inner_angle>0.9</inner_angle>
<outer_angle>1.0</outer_angle>
<falloff>2.0</falloff>
```

```
</spot>

<direction>0 0 -1</direction>

<cast_shadows>1</cast_shadows>

</light>

</link>

<joint name="warnign_light_joint" type="revolute">

<child>warnign_light</child>

<parent>base_link</parent>

<axis>

<xyz>0 0 1</xyz>

<limit>

<effort>1000</effort>

<velocity>20.0</velocity>

</limit>

</axis>

</joint>

<!-- SENSORS -->

<link name="camera_front">

<pose relative_to="base_link">0.0553 0 0.4323 0 0 0</pose>

<sensor name="color" type="camera">

<pose>0 0.0325 0.0125 0 0 0</pose>

<update_rate>6</update_rate>

<always_on>0</always_on>

<camera>

<horizontal_fov>1.20427718</horizontal_fov>

<image>

<width>848</width>

<height>480</height>

</image>
```

```
<clip>
<near>0.1</near>
<far>50</far>
</clip>
<lens>
<intrinsics>
<fx>615.9603271484375</fx>
<fy>616.227294921875</fy>
<cx>419.83026123046875</cx>
<cy>245.1431427001953</cy>
<s>0</s>
</intrinsics>
</lens>
</camera>
</sensor>
<sensor name="depth" type="depth_camera">
<pose>0 0.0175 0.0125 0 0 0</pose>
<update_rate>6</update_rate>
<always_on>0</always_on>
<camera>
<horizontal_fov>1.51843645</horizontal_fov>
<image>
<width>848</width>
<height>480</height>
<format>R_FLOAT32</format>
</image>
<clip>
<near>0.1</near>
<far>10</far>
```

```
</clip>

<lens>

<intrinsics>

<fx>421.61578369140625</fx>

<fy>421.61578369140625</fy>

<cx>422.2854309082031</cx>

<cy>236.57237243652344</cy>

<s>0</s>

</intrinsics>

</lens>

</camera>

</sensor>

</link>

<joint name="camera_front_joint" type="fixed">

<child>camera_front</child>

<parent>base_link</parent>

</joint>

<link name="camera_back">

<pose relative_to="base_link">-0.241 0 0.2303 0 0 3.141592654</pose>

<sensor name="color" type="camera">

<pose>0 0.0325 0.0125 0 0 0</pose>

<update_rate>6</update_rate>

<always_on>0</always_on>

<camera>

<horizontal_fov>1.20427718</horizontal_fov>

<image>

<width>848</width>

<height>480</height>

</image>
```

```xml
<clip>
<near>0.1</near>
<far>50</far>
</clip>
<lens>
<intrinsics>
<fx>615.9603271484375</fx>
<fy>616.227294921875</fy>
<cx>419.83026123046875</cx>
<cy>245.1431427001953</cy>
<s>0</s>
</intrinsics>
</lens>
</camera>
</sensor>
<sensor name="depth" type="depth_camera">
<pose>0 0.0175 0.0125 0 0 0</pose>
<update_rate>6</update_rate>
<always_on>0</always_on>
<camera>
<horizontal_fov>1.51843645</horizontal_fov>
<image>
<width>848</width>
<height>480</height>
<format>R_FLOAT32</format>
</image>
<clip>
<near>0.1</near>
<far>10</far>
```

```xml
</clip>
<lens>
<intrinsics>
<fx>421.61578369140625</fx>
<fy>421.61578369140625</fy>
<cx>422.2854309082031</cx>
<cy>236.57237243652344</cy>
<s>0</s>
</intrinsics>
</lens>
</camera>
</sensor>
</link>
<joint name="camera_back_joint" type="fixed">
<child>camera_back</child>
<parent>base_link</parent>
</joint>
<link name="scan_front">
<pose>0.221 0 0.1404 0 0 0</pose>
<sensor name="scan_front" type="gpu_lidar">
<update_rate>10</update_rate>
<lidar>
<scan>
<horizontal>
<samples>674</samples>
<resolution>1</resolution>
<min_angle>-1.47043991089</min_angle>
<max_angle>1.47043991089</max_angle>
</horizontal>
```

```
</scan>

<range>

<min>0.01</min>

<max>5.0</max>

<resolution>0.01</resolution>

</range>

</lidar>

</sensor>

</link>

<joint name="scan_front_joint" type="fixed">

<child>scan_front</child>

<parent>base_link</parent>

</joint>

<link name="scan_back">

<pose>-0.2075 0 0.205 0 0 3.141592654</pose>

<sensor name="scan_back" type="gpu_lidar">

<update_rate>5</update_rate>

<lidar>

<scan>

<horizontal>

<samples>480</samples>

<resolution>1</resolution>

<min_angle>-1.47043991089</min_angle>

<max_angle>1.47043991089</max_angle>

</horizontal>

</scan>

<range>

<min>0.01</min>

<max>5.0</max>
```

```
<resolution>0.01</resolution>
</range>
</lidar>
</sensor>
</link>
<joint name="scan_back_joint" type="fixed">
<child>scan_back</child>
<parent>base_link</parent>
</joint>
<link name="scan_omni">
<pose>-0.1855 0 0.5318 0 0 0</pose>
<visual name="velodyne_base">
<pose>0 0 -0.035 0 0 0</pose>
<geometry>
<cylinder>
<radius>0.06</radius>
<length>0.0015</length>
</cylinder>
</geometry>
</visual>
<visual name="velodyne_suport_1">
<pose>-0.035 0.035 -0.08 0 0 0</pose>
<geometry>
<cylinder>
<radius>0.005</radius>
<length>0.09</length>
</cylinder>
</geometry>
</visual>
```

```
<visual name="velodyne_suport_2">

<pose>-0.035 -0.035 -0.08 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.005</radius>

<length>0.09</length>

</cylinder>

</geometry>

</visual>

<visual name="velodyne_suport_3">

<pose>0.0495 0 -0.08 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.005</radius>

<length>0.09</length>

</cylinder>

</geometry>

</visual>

<!-- Bottom cylinder of Ouster/Velodyne 3D Lidar -->

<visual name="velodyne_base_link_visual_base">

<pose>0 0 -0.035 0 0 0</pose>

<geometry>

<mesh>

<scale>1 1 1</scale>

<uri>meshes/VLP16_base_1.dae</uri>

</mesh>

</geometry>

</visual>

<!-- Top cylinder of Ouster/Velodyne 3D Lidar -->
```

```xml
<visual name="velodyne_base_link_visual_sensor">

<pose>0 0 -0.035 0 0 0</pose>

<geometry>

<mesh>

<scale>1 1 1</scale>

<uri>meshes/VLP16_base_2.dae</uri>

</mesh>

</geometry>

</visual>

<!-- Main cylinder of Ouster/Velodyne 3D Lidar -->

<visual name="base_link_velodyne_visual_scan">

<pose>0 0 -0.035 0 0 0</pose>

<geometry>

<mesh>

<scale>1 1 1</scale>

<uri>meshes/VLP16_scan.dae</uri>

</mesh>

</geometry>

</visual>

<sensor name="scan_omni" type="gpu_lidar">

<update_rate>10</update_rate>

<lidar>

<scan>

<horizontal>

<!-- Real samples value is 1800 -->

<samples>900</samples>

<resolution>1</resolution>

<min_angle>-3.141592654</min_angle>

<max_angle>3.141592654</max_angle>
```

```
</horizontal>
<vertical>
<samples>16</samples>
<resolution>1</resolution>
<min_angle>-0.261799388</min_angle>
<max_angle>0.261799388</max_angle>
</vertical>
</scan>
<range>
<min>0.2</min>
<max>100.0</max>
<resolution>0.01</resolution>
</range>
</lidar>
</sensor>
</link>
<joint name="scan_omni_joint" type="fixed">
<child>scan_omni</child>
<parent>base_link</parent>
</joint>
<!-- GRIPPER -->
<link name="gripper">
<pose>-0.35 0 0.1 0 0 0</pose>
<inertial>
<pose>0 0 0 0 -0 0</pose>
<mass>1</mass>
<inertia>
<ixx>0.001666667</ixx>
<ixy>0</ixy>
```

<ixz>0</ixz>

<iyy>0.001666667</iyy>

<iyz>0</iyz>

<izz>0.001666667</izz>

</inertia>

</inertial>

<collision name="gripper_collision">

<pose>-0.005 0 0 0 -0 0</pose>

<max_contacts>5</max_contacts>

<geometry>

<box>

<size>0.15 0.1 0.1</size>

</box>

</geometry>

<surface>

<contact>

<ode>

<kp>1000000.0</kp>

<kd>1</kd>

<min_depth>0.01</min_depth>

<max_vel>1</max_vel>

</ode>

</contact>

<friction>

<ode>

<mu>1.16</mu>

<mu2>1.16</mu2>

</ode>

</friction>

```xml
</surface>

</collision>

<visual name="gripper_visual">

<pose>0 0 0 -1.57079632679 0 0</pose>

<geometry>

<mesh>

<scale>1 1 1</scale>

<uri>meshes/gripper2/gripper2.dae</uri>

</mesh>

</geometry>

</visual>

<sensor name="sensor_contact" type="gpu_lidar">

<pose>-0.075 0 0.035 0 0 3.141592654</pose>

<update_rate>10</update_rate>

<lidar>

<scan>

<horizontal>

<samples>3</samples>

<resolution>1</resolution>

<min_angle>-0.02</min_angle>

<max_angle>0.02</max_angle>

</horizontal>

</scan>

<range>

<min>0.001</min>

<max>0.01</max>

<resolution>0.001</resolution>

</range>

</lidar>
```

```xml
</sensor>
</link>
<joint name="gripper_joint" type="revolute">
<pose>0.35 0 0.09 0 0 0</pose>
<child>gripper</child>
<parent>base_link</parent>
<axis>
<xyz>0 0 1</xyz>
<limit>
<lower>-0.7</lower>
<upper>0.7</upper>
<effort>4</effort>
<velocity>0.5</velocity>
</limit>
<dynamics>
<friction>0.5</friction>
</dynamics>
</axis>
</joint>
<link name="gripper_hand">
<pose>-0.463 0 0.1 0 0 0</pose>
<inertial>
<pose>0 0 0 0 0 0</pose>
<mass>0.2</mass>
<inertia>
<ixx>0.000016</ixx>
<ixy>0</ixy>
<ixz>0</ixz>
<iyy>0.000016</iyy>
```

```
<iyz>0</iyz>

<izz>0.000016</izz>

</inertia>

</inertial>

<collision name="gripper_hand_collision">

<pose>0.015 0 0.006 0 0 0</pose>

<geometry>

<box>

<size>0.04 0.1 0.017</size>

</box>

</geometry>

<max_contacts>5</max_contacts>

<surface>

<contact>

<ode>

<kp>1000000.0</kp>

<kd>1</kd>

<min_depth>0.01</min_depth>

<max_vel>1</max_vel>

</ode>

</contact>

<friction>

<ode>

<mu>1.16</mu>

<mu2>1.16</mu2>

</ode>

</friction>

</surface>

</collision>
```

```
<visual name="gripper_hand_visual">

<pose>0 0 0 1.570796327 0 1.570796327</pose>

<geometry>

<mesh>

<scale>0.001 0.001 0.001</scale>

<uri>meshes/gripper2/gripper_hand.stl</uri>

</mesh>

</geometry>

<material>

<ambient>0.1 0.1 0.1 1</ambient>

<diffuse>0.3 0.3 0.3 1</diffuse>

<specular>1 1 1 1</specular>

<emissive>0 0 0 1</emissive>

</material>

</visual>

<collision name="gripper_hand_sideL_collision">

<pose>0 0.08 0.03 0 0 0</pose>

<geometry>

<box>

<size>0.01 0.05 0.07</size>

</box>

</geometry>

<surface>

<contact>

<ode>

<kp>1000000.0</kp>

<kd>1</kd>

<min_depth>0.01</min_depth>

<max_vel>1</max_vel>
```

</ode>

</contact>

<friction>

<ode>

<mu>1.16</mu>

<mu2>1.16</mu2>

</ode>

</friction>

</surface>

</collision>

<collision name="gripper_hand_sideL_slide_collision">

<pose>0 0.055 0.03 0.9 0 0</pose>

<geometry>

<box>

<size>0.01 0.055 0.04</size>

</box>

</geometry>

<surface>

<contact>

<ode>

<kp>1000000.0</kp>

<kd>1</kd>

<min_depth>0.01</min_depth>

<max_vel>1</max_vel>

</ode>

</contact>

<friction>

<ode>

<mu>0</mu>

```
<mu2>0</mu2>
</ode>
</friction>
</surface>
</collision>
<collision name="gripper_hand_sideR_collision">
<pose>0 -0.08 0.03 0 0 0</pose>
<geometry>
<box>
<size>0.01 0.05 0.07</size>
</box>
</geometry>
<surface>
<contact>
<ode>
<kp>1000000.0</kp>
<kd>1</kd>
<min_depth>0.01</min_depth>
<max_vel>1</max_vel>
</ode>
</contact>
<friction>
<ode>
<mu>1.16</mu>
<mu2>1.16</mu2>
</ode>
</friction>
</surface>
</collision>
```

```xml
<collision name="gripper_hand_sideR_slide_collision">
<pose>0 -0.055 0.03 -0.9 0 0</pose>
<geometry>
<box>
<size>0.01 0.055 0.04</size>
</box>
</geometry>
<surface>
<contact>
<ode>
<kp>1000000.0</kp>
<kd>1</kd>
<min_depth>0.01</min_depth>
<max_vel>1</max_vel>
</ode>
</contact>
<friction>
<ode>
<mu>0</mu>
<mu2>0</mu2>
</ode>
</friction>
</surface>
</collision>
</link>
<joint name="gripper_hand_joint" type="revolute">
<pose>0.2 0 0 0 0 0</pose>
<child>gripper_hand</child>
<parent>gripper</parent>
```

```xml
<axis>
<xyz>0 1 0</xyz>
<limit>
<lower>-0.35</lower>
<upper>0.1</upper>
<effort>100</effort>
<velocity>0.5</velocity>
</limit>
</axis>
</joint>
<!-- WHEELS -->
<link name="wheel_front">
<pose>0.145 0 0.0345 1.570796327 0 0</pose>
<inertial>
<pose>0 0 0 0 0 0</pose>
<mass>0.1</mass>
<inertia>
<ixx>0.0000175</ixx>
<ixy>0</ixy>
<ixz>0</ixz>
<iyy>0.0000175</iyy>
<iyz>0</iyz>
<izz>0.0000175</izz>
</inertia>
</inertial>
<collision name="wheel_front_collision">
<pose>0 0 0 0 0 0</pose>
<geometry>
<sphere>
```

```xml
<radius>0.025</radius>
</sphere>
</geometry>
<max_contacts>2</max_contacts>
<surface>
<contact>
<ode>
<kp>1000000.0</kp>
<kd>1</kd>
<min_depth>0.01</min_depth>
<max_vel>1</max_vel>
</ode>
</contact>
<friction>
<ode>
<mu>1.16</mu>
<mu2>1.16</mu2>
</ode>
</friction>
</surface>
</collision>
</link>
<joint name="wheel_front_joint" type="ball">
<child>wheel_front</child>
<parent>base_link</parent>
<axis>
<limit>
<effort>100</effort>
</limit>
```

```
<dynamics>
<spring_reference>0</spring_reference>
<spring_stiffness>0</spring_stiffness>
<damping>10</damping>
</dynamics>
</axis>
</joint>
<link name="wheel_back">
<pose>-0.225 0 0.035 1.570796327 0 0</pose>
<inertial>
<pose>0 0 0 0 0 0</pose>
<mass>0.1</mass>
<inertia>
<ixx>0.0000175</ixx>
<ixy>0</ixy>
<ixz>0</ixz>
<iyy>0.0000175</iyy>
<iyz>0</iyz>
<izz>0.0000175</izz>
</inertia>
</inertial>
<collision name="wheel_back_collision">
<pose>0 0 0 0 0 0</pose>
<geometry>
<sphere>
<radius>0.035</radius>
</sphere>
</geometry>
<max_contacts>2</max_contacts>
```

```
<surface>
<contact>
<ode>
<kp>1000000.0</kp>
<kd>1</kd>
<min_depth>0.01</min_depth>
<max_vel>1</max_vel>
</ode>
</contact>
<friction>
<ode>
<mu>1.16</mu>
<mu2>1.16</mu2>
</ode>
</friction>
</surface>
</collision>
</link>
<joint name="wheel_back_joint" type="ball">
<child>wheel_back</child>
<parent>base_link</parent>
<axis>
<limit>
<effort>100</effort>
</limit>
<dynamics>
<spring_reference>0</spring_reference>
<spring_stiffness>0</spring_stiffness>
<damping>10</damping>
```

</dynamics>

</axis>

</joint>

<link name="wheel_left">

<pose>0 0.257 0.195 -1.570796327 0 0</pose>

<inertial>

<pose>0 0 0 0 0 0</pose>

<mass>0.1</mass>

<inertia>

<ixx>0.0000175</ixx>

<ixy>0</ixy>

<ixz>0</ixz>

<iyy>0.0000175</iyy>

<iyz>0</iyz>

<izz>0.00003125</izz>

</inertia>

</inertial>

<collision name="wheel_left_collision">

<pose>0 0 0 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.195</radius>

<length>0.05</length>

</cylinder>

</geometry>

<max_contacts>5</max_contacts>

<surface>

<contact>

<ode>

```
<kp>1000000.0</kp>
<kd>1</kd>
<min_depth>0.01</min_depth>
<max_vel>1</max_vel>
</ode>
</contact>
<friction>
<ode>
<mu>1.16</mu>
<mu2>10</mu2>
<fdir1>1 0 0</fdir1>
</ode>
</friction>
</surface>
</collision>
<visual name="wheel_left_visual">
<pose>0 0 -0.03 0 0 0</pose>
<geometry>
<mesh>
<scale>0.98 0.98 1.0</scale>
<uri>meshes/wheel/wheel.dae</uri>
</mesh>
</geometry>
</visual>
</link>
<joint name="wheel_left_joint" type="revolute">
<child>wheel_left</child>
<parent>base_link</parent>
<axis>
```

```
<xyz>0 0 1</xyz>
<limit>
<effort>9.6</effort>
</limit>
<dynamics>
<damping>3.0</damping>
<friction>0.5</friction>
</dynamics>
</axis>
</joint>
<link name="wheel_right">
<pose>0 -0.257 0.195 1.570796327 0 0</pose>
<inertial>
<pose>0 0 0 0 0 0</pose>
<mass>0.1</mass>
<inertia>
<ixx>0.0000175</ixx>
<ixy>0</ixy>
<ixz>0</ixz>
<iyy>0.0000175</iyy>
<iyz>0</iyz>
<izz>0.00003125</izz>
</inertia>
</inertial>
<collision name="wheel_right_collision">
<pose>0 0 0 0 0 0</pose>
<geometry>
<cylinder>
<radius>0.195</radius>
```

```
<length>0.05</length>
</cylinder>
</geometry>
<max_contacts>5</max_contacts>
<surface>
<contact>
<ode>
<kp>1000000.0</kp>
<kd>1</kd>
<min_depth>0.01</min_depth>
<max_vel>1</max_vel>
</ode>
</contact>
<friction>
<ode>
<mu>1.16</mu>
<mu2>10</mu2>
<fdir1>1 0 0</fdir1>
</ode>
</friction>
</surface>
</collision>
<visual name="wheel_right_visual">
<pose>0 0 -0.03 0 0 0</pose>
<geometry>
<mesh>
<scale>0.98 0.98 1.0</scale>
<uri>meshes/wheel/wheel.dae</uri>
</mesh>
```

```xml
</geometry>
</visual>
</link>
<joint name="wheel_right_joint" type="revolute">
<child>wheel_right</child>
<parent>base_link</parent>
<axis>
<xyz>0 0 -1</xyz>
<limit>
<effort>9.6</effort>
</limit>
<dynamics>
<damping>3.0</damping>
<friction>0.5</friction>
</dynamics>
</axis>
</joint>
<!-- CONTROLERS PLUGINS -->
<!-- DIFF DRIVE PLUGIN / WHEELS CONTROLER -->
<plugin filename="ignition-gazebo-diff-drive-system" name="ignition::gazebo::systems::DiffDrive">
<left_joint>wheel_left_joint</left_joint>
<right_joint>wheel_right_joint</right_joint>
<wheel_separation>0.5605</wheel_separation>
<wheel_radius>0.195</wheel_radius>
<odom_publish_frequency>20</odom_publish_frequency>
<frame_id>odom</frame_id>
<child_frame_id>base_link</child_frame_id>
</plugin>
<!-- GRIPPER RAIL CONTROLER -->
```

```
<plugin filename="ignition-gazebo-joint-controller-system" name="ignition::gazebo::systems::JointController">

<joint_name>gripper_joint</joint_name>

<use_force_commands>true</use_force_commands>

<p_gain>0.2</p_gain>

<i_gain>0.01</i_gain>

</plugin>

<!-- GRIPPER ATTACH CONTROLER -->

<plugin filename="ignition-gazebo-joint-controller-system" name="ignition::gazebo::systems::JointController">

<joint_name>gripper_hand_joint</joint_name>

<use_force_commands>true</use_force_commands>

<p_gain>0.2</p_gain>

<i_gain>0.01</i_gain>

</plugin>

<!-- GROUND ODOMETRY POSITION STATE PUBLISHER (FAKE LOCALIZATION) -->

<plugin filename="ignition-gazebo-pose-publisher-system" name="ignition::gazebo::systems::PosePublisher">

<publish_link_pose>true</publish_link_pose>

<publish_collision_pose>false</publish_collision_pose>

<publish_visual_pose>false</publish_visual_pose>

<publish_sensor_pose>false</publish_sensor_pose>

<publish_nested_model_pose>true</publish_nested_model_pose>

<update_frequency>20</update_frequency>

</plugin>

<plugin filename="ignition-gazebo-joint-controller-system" name="ignition::gazebo::systems::JointController">

<joint_name>warnign_light_joint</joint_name>

<initial_velocity>10.0</initial_velocity>

</plugin>

<plugin filename="ignition-gazebo-linearbatteryplugin-system"
name="ignition::gazebo::systems::LinearBatteryPlugin">

<battery_name>linear_battery</battery_name>

<voltage>24.592</voltage>
```

```
<open_circuit_voltage_constant_coef>24.592</open_circuit_voltage_constant_coef>

<capacity>1.2009</capacity>

<power_load>6.5</power_load>

<fix_issue_225>true</fix_issue_225>

<enable_recharge>true</enable_recharge>

<!-- charging I = c / t, discharging I = P / V,

charging I should be > discharging I -->

<charging_time>3.0</charging_time>

<!-- Consumer-specific -->

<!-- <power_load>2.1</power_load> -->

<recharge_by_topic>true</recharge_by_topic>

</plugin>

<plugin filename="ignition-gazebo-joint-state-publisher-system"
name="ignition::gazebo::systems::JointStatePublisher">

<joint_name>gripper_joint</joint_name>

<joint_name>gripper_hand_joint</joint_name>

</plugin>

</model>

</sdf>)model.urdf(<?xml version="1.0" ?>

<robot name="tugbot">

        <link name="base_link">

                <origin rpy="0 0 0" xyz="0 0 0"/>

                <visual>

                        <origin rpy="0 0 0" xyz="0 0 0"/>

                        <geometry>

                                <mesh
filename="package://tugbot_description/meshes/base/tugbot_simp.dae"/>

                        </geometry>

                </visual>

                <visual>
```

```
				<origin rpy="0 0 0" xyz="0 0 0"/>
				<geometry>
					<mesh
filename="package://tugbot_description/meshes/base/movai_logo.dae"/>
				</geometry>
				<material name="">
					<color rgba="1.0 1.0 1.0 1.0"/>
				</material>
			</visual>
			<visual>
				<origin rpy="0 1.57079632679 0" xyz="-0.1 0 0.1945"/>
				<geometry>
					<mesh
filename="package://tugbot_description/meshes/base/movai-logo.png"/>
				</geometry>
				<material name="">
					<color rgba="0.0 0.0 0 1"/>
				</material>
			</visual>
			<visual>
				<origin rpy="0 1.57079632679 0" xyz="-0.099 0 0.195"/>
				<geometry>
					<mesh filename="file://"/>
				</geometry>
				<material name="">
					<color rgba="1 0.5 0 0.5"/>
				</material>
			</visual>
		</link>
		<link name="imu_link">
```

```
		<origin rpy="0 0 -1.57" xyz="0.14 0.02 0.25"/>
	</link>
	<joint name="imu_link_joint" type="fixed">
		<origin rpy="0 0 -1.57" xyz="0.14 0.02 0.25"/>
		<child link="imu_link"/>
		<parent link="base_link"/>
	</joint>
	<link name="warnign_light">
		<origin rpy="0 0 0" xyz="-0.185 0 0.46"/>
		<inertial>
			<origin rpy="0 0 0" xyz="0 0 0"/>
			<mass value="0.1"/>
			<inertia ixx="0.000016875" ixy="0" ixz="0" iyy="0.00000001" iyz="0"
izz="0.000016875"/>
		</inertial>
	</link>
	<joint name="warnign_light_joint" type="fixed">
		<origin rpy="0 0 0" xyz="-0.185 0 0.46"/>
		<child link="warnign_light"/>
		<parent link="base_link"/>
		<axis xyz="0 0 1"/>
		<limit effort="1000" velocity="20.0"/>
	</joint>
	<link name="camera_front">
		<origin rpy="0 0 0" xyz="0.0553 0 0.4323"/>
	</link>
	<joint name="camera_front_joint" type="fixed">
		<origin rpy="0 0 0" xyz="0.0553 0 0.4323"/>
		<child link="camera_front"/>
		<parent link="base_link"/>
```

```
</joint>
<link name="camera_back">
        <origin rpy="0 0 3.141592654" xyz="-0.241 0 0.2303"/>
</link>
<joint name="camera_back_joint" type="fixed">
        <origin rpy="0 0 3.141592654" xyz="-0.241 0 0.2303"/>
        <child link="camera_back"/>
        <parent link="base_link"/>
</joint>
<link name="tugbot/scan_front/scan_front">
        <origin rpy="0 0 0" xyz="0.221 0 0.1404"/>
</link>
<joint name="scan_front_joint" type="fixed">
        <origin rpy="0 0 0" xyz="0.221 0 0.1404"/>
        <child link="tugbot/scan_front/scan_front"/>
        <parent link="base_link"/>
</joint>
<link name="tugbot/scan_back/scan_back">
        <origin rpy="0 0 3.141592654" xyz="-0.2075 0 0.205"/>
</link>
<joint name="scan_back_joint" type="fixed">
        <origin rpy="0 0 3.141592654" xyz="-0.2075 0 0.205"/>
        <child link="tugbot/scan_back/scan_back"/>
        <parent link="base_link"/>
</joint>
<link name="tugbot/scan_omni/scan_omni">
        <origin rpy="0 0 0" xyz="-0.1855 0 0.5318"/>
        <visual>
                <origin rpy="0 0 0" xyz="0 0 -0.035"/>
```

```
        <geometry>
            <cylinder radius="0.06" length="0.0015"/>
        </geometry>
    </visual>
    <visual>
        <origin rpy="0 0 0" xyz="-0.035 0.035 -0.08"/>
        <geometry>
            <cylinder radius="0.005" length="0.09"/>
        </geometry>
    </visual>
    <visual>
        <origin rpy="0 0 0" xyz="-0.035 -0.035 -0.08"/>
        <geometry>
            <cylinder radius="0.005" length="0.09"/>
        </geometry>
    </visual>
    <visual>
        <origin rpy="0 0 0" xyz="0.0495 0 -0.08"/>
        <geometry>
            <cylinder radius="0.005" length="0.09"/>
        </geometry>
    </visual>
    <visual>
        <origin rpy="0 0 0" xyz="0 0 -0.035"/>
        <geometry>
            <mesh
filename="package://tugbot_description/meshes/VLP16_base_1.dae"/>
        </geometry>
    </visual>
    <visual>
```

```
<origin rpy="0 0 0" xyz="0 0 -0.035"/>
<geometry>
<mesh
filename="package://tugbot_description/meshes/VLP16_base_2.dae"/>
</geometry>
</visual>
<visual>
<origin rpy="0 0 0" xyz="0 0 -0.035"/>
<geometry>
<mesh filename="package://tugbot_description/meshes/VLP16_scan.dae"/>
</geometry>
</visual>
</link>
<joint name="scan_omni_joint" type="fixed">
<origin rpy="0 0 0" xyz="-0.1855 0 0.5318"/>
<child link="tugbot/scan_omni/scan_omni"/>
<parent link="base_link"/>
</joint>
<!-- <link name="wheel_front">
<origin rpy="1.570796327 0 0" xyz="0.145 0 0.0345"/>
<inertial>
<origin rpy="0 0 0" xyz="0 0 0"/>
<mass value="0.1"/>
<inertia ixx="0.0000175" ixy="0" ixz="0" iyy="0.0000175" iyz="0" izz="0.0000175"/>
</inertial>
<collision>
<origin rpy="0 0 0" xyz="0 0 0"/>
<geometry>
<sphere radius="0.025"/>
</geometry>
```

```
		</collision>
	</link>
	<link name="wheel_back">
		<origin rpy="1.570796327 0 0" xyz="-0.225 0 0.035"/>
		<inertial>
			<origin rpy="0 0 0" xyz="0 0 0"/>
			<mass value="0.1"/>
			<inertia ixx="0.0000175" ixy="0" ixz="0" iyy="0.0000175" iyz="0" izz="0.0000175"/>
		</inertial>
		<collision>
			<origin rpy="0 0 0" xyz="0 0 0"/>
			<geometry>
				<sphere radius="0.035"/>
			</geometry>
		</collision>
	</link> -->
	<link name="wheel_left">
		<origin rpy="-1.570796327 0 0" xyz="0 0.257 0.195"/>
		<inertial>
			<origin rpy="0 0 0" xyz="0 0 0"/>
			<mass value="0.1"/>
			<inertia ixx="0.0000175" ixy="0" ixz="0" iyy="0.0000175" iyz="0" izz="0.00003125"/>
		</inertial>
		<collision>
			<origin rpy="0 0 0" xyz="0 0 0"/>
			<geometry>
				<cylinder radius="0.195" length="0.05"/>
			</geometry>
		</collision>
```

```
        <visual>
            <origin rpy="0 0 0" xyz="0 0 -0.03"/>
            <geometry>
                <mesh filename="package://tugbot_description/meshes/wheel/wheel.dae"/>
            </geometry>
        </visual>
    </link>
    <joint name="wheel_left_joint" type="fixed">
        <origin rpy="-1.570796327 0 0" xyz="0 0.257 0.195"/>
        <child link="wheel_left"/>
        <parent link="base_link"/>
        <axis xyz="0 0 1"/>
<limit effort="9.6" velocity="1.0"/>
        <dynamics damping="3.0" friction="0.5"/>
    </joint>
    <link name="wheel_right">
        <origin rpy="1.570796327 0 0" xyz="0 -0.257 0.195"/>
        <inertial>
            <origin rpy="0 0 0" xyz="0 0 0"/>
            <mass value="0.1"/>
            <inertia ixx="0.0000175" ixy="0" ixz="0" iyy="0.0000175" iyz="0" izz="0.00003125"/>
        </inertial>
        <collision>
            <origin rpy="0 0 0" xyz="0 0 0"/>
            <geometry>
                <cylinder radius="0.195" length="0.05"/>
            </geometry>
        </collision>
        <visual>
```

```
                <origin rpy="0 0 0" xyz="0 0 -0.03"/>
                <geometry>
                    <mesh filename="package://tugbot_description/meshes/wheel/wheel.dae"/>
                </geometry>
            </visual>
        </link>
        <joint name="wheel_right_joint" type="fixed">
            <origin rpy="1.570796327 0 0" xyz="0 -0.257 0.195"/>
            <child link="wheel_right"/>
            <parent link="base_link"/>
            <axis xyz="0 0 -1"/>
<limit effort="9.6" velocity="1.0"/>
            <dynamics damping="3.0" friction="0.5"/>
        </joint>
</robot>)rviz.rviz(Panels:
- Class: rviz_common/Displays
Help Height: 78
Name: Displays
Property Tree Widget:
Expanded:
- /Global Options1
- /Status1
- /TF1
- /TF1/Frames1
- /TF1/Tree1
- /RobotModel1
- /RobotModel1/Description Topic1
Splitter Ratio: 0.5
Tree Height: 839
```

- Class: rviz_common/Selection

Name: Selection

- Class: rviz_common/Tool Properties

Expanded:

- /2D Goal Pose1

- /Publish Point1

Name: Tool Properties

Splitter Ratio: 0.5886790156364441

- Class: rviz_common/Views

Expanded:

- /Current View1

Name: Views

Splitter Ratio: 0.5

- Class: rviz_common/Time

Experimental: false

Name: Time

SyncMode: 0

SyncSource: ""

Visualization Manager:

Class: ""

Displays:

- Alpha: 0.5

Cell Size: 1

Class: rviz_default_plugins/Grid

Color: 160; 160; 164

Enabled: true

Line Style:

Line Width: 0.029999999329447746

Value: Lines

Name: Grid

Normal Cell Count: 0

Offset:

X: 0

Y: 0

Z: 0

Plane: XY

Plane Cell Count: 10

Reference Frame: <Fixed Frame>

Value: true

- Class: rviz_default_plugins/TF

Enabled: true

Frame Timeout: 15

Frames:

All Enabled: true

base_link:

Value: true

camera_back:

Value: true

camera_front:

Value: true

imu_link:

Value: true

scan_back:

Value: true

scan_front:

Value: true

scan_omni:

Value: true

warnign_light:

Value: true

wheel_left:

Value: true

wheel_right:

Value: true

Marker Scale: 1

Name: TF

Show Arrows: true

Show Axes: true

Show Names: true

Tree:

base_link:

camera_back:

{}

camera_front:

{}

imu_link:

{}

scan_back:

{}

scan_front:

{}

scan_omni:

{}

warnign_light:

{}

wheel_left:

{}

wheel_right:

{}

Update Interval: 0

Value: true

- Alpha: 1

Class: rviz_default_plugins/RobotModel

Collision Enabled: false

Description File: ""

Description Source: Topic

Description Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /robot_description

Enabled: true

Links:

All Links Enabled: true

Expand Joint Details: false

Expand Link Details: false

Expand Tree: false

Link Tree Style: Links in Alphabetic Order

base_link:

Alpha: 1

Show Axes: false

Show Trail: false

Value: true

camera_back:

Alpha: 1

Show Axes: false

Show Trail: false

camera_front:

Alpha: 1

Show Axes: false

Show Trail: false

imu_link:

Alpha: 1

Show Axes: false

Show Trail: false

scan_back:

Alpha: 1

Show Axes: false

Show Trail: false

scan_front:

Alpha: 1

Show Axes: false

Show Trail: false

scan_omni:

Alpha: 1

Show Axes: false

Show Trail: false

Value: true

warnign_light:

Alpha: 1

Show Axes: false

Show Trail: false

wheel_left:

Alpha: 1

Show Axes: false

Show Trail: false

Value: true

wheel_right:

Alpha: 1

Show Axes: false

Show Trail: false

Value: true

Mass Properties:

Inertia: false

Mass: false

Name: RobotModel

TF Prefix: ""

Update Interval: 0

Value: true

Visual Enabled: true

Enabled: true

Global Options:

Background Color: 48; 48; 48

Fixed Frame: base_link

Frame Rate: 30

Name: root

Tools:

- Class: rviz_default_plugins/Interact

Hide Inactive Objects: true

- Class: rviz_default_plugins/MoveCamera

- Class: rviz_default_plugins/Select

- Class: rviz_default_plugins/FocusCamera

- Class: rviz_default_plugins/Measure

Line color: 128; 128; 0

- Class: rviz_default_plugins/SetInitialPose

Covariance x: 0.25

Covariance y: 0.25

Covariance yaw: 0.06853891909122467

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /initialpose

- Class: rviz_default_plugins/SetGoal

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /goal_pose

- Class: rviz_default_plugins/PublishPoint

Single click: true

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /clicked_point

Transformation:

Current:

Class: rviz_default_plugins/TF

Value: true

Views:

Current:

Class: rviz_default_plugins/Orbit

Distance: 1.1108241081237793

Enable Stereo Rendering:

Stereo Eye Separation: 0.05999999865889549

Stereo Focal Distance: 1

Swap Stereo Eyes: false

Value: false

Focal Point:

X: -0.04430290684103966

Y: 0.0008574090898036957

Z: 0.22932319343090057

Focal Shape Fixed Size: true

Focal Shape Size: 0.05000000074505806

Invert Z Axis: false

Name: Current View

Near Clip Distance: 0.009999999776482582

Pitch: 0.4703982472419739

Target Frame: <Fixed Frame>

Value: Orbit (rviz)

Yaw: 0.7503951787948608

Saved: ~

Window Geometry:

Displays:

collapsed: false

Height: 1136

Hide Left Dock: false

Hide Right Dock: false

QMainWindow State:
000000ff00000000fd000000040000000000000156000003d2fc0200000008fb0000001200530065006c00650063
00740069006f006e00000001e10000009b0000005c00fffffffb0000001e0054006f006f006c002000500072006f0070
006500720074006900650073020000001ed000001df00000185000000a3fb0000001200560069006500770073002
00054006f006f02000001df000002110000018500000122fb000000200054006f006f006c002000500072006f0070
006500720074006900650073003203000002880000011d000002210000017afb000000100044006900730070000
6c006100790073010000003d000003d2000000c900fffffffb0000002000730065006c0065006300740069006f006e
00200062007500660066006500720200000138000000aa0000023a00000294fb00000014005700690064006500
530074006500720065006f02000000e6000000d2000003ee0000030bfb0000000c004b0069006e0065006300740
200000186000001060000030c00000261000000010000010f000003d2fc0200000003fb0000001e0054006f006f0
06c002000500072006f00700065007200740069006500730100000041000000780000000000000000fb0000000a
00560069006500770073010000003d000003d2000000a400fffffffb0000001200530065006c006500630074006900
6f006e010000025a000000b200000000000000000000000200000490000000a9fc0100000001fb0000000a00560
069006500770073030000004e00000080000002e100000197000000030000073a0000003efc0100000002fb0000
000800540069006d006501000000000000073a000002fb00fffffffb0000000800540069006d0065010000000000000
0450000000000000000000004c9000003d200000004000000040000000800000008fc00000001000000020000
00010000000a0054006f006f006c00730100000000ffffffff0000000000000000

Selection:

collapsed: false

Time:

collapsed: false

Tool Properties:

collapsed: false

Views:

collapsed: false

Width: 1850

X: 70

Y: 27)il y a aussi toujours dans tugbot_description CMaleLists.txt (cmake_minimum_required(VERSION 3.8)

project(tugbot_description)

if(CMAKE_COMPILER_IS_GNUCXX OR CMAKE_CXX_COMPILER_ID MATCHES "Clang")

add_compile_options(-Wall -Wextra -Wpedantic)

endif()

find dependencies

find_package(ament_cmake REQUIRED)

find_package(robot_state_publisher REQUIRED)

find_package(xacro REQUIRED)

```
if(BUILD_TESTING)
find_package(ament_lint_auto REQUIRED)
# the following line skips the linter which checks for copyrights
# comment the line when a copyright and license is added to all source files
set(ament_cmake_copyright_FOUND TRUE)
# the following line skips cpplint (only works in a git repo)
# comment the line when this package is in a git repo and when
# a copyright and license is added to all source files
set(ament_cmake_cpplint_FOUND TRUE)
ament_lint_auto_find_test_dependencies()
endif()
install(DIRECTORY
launch
meshes
urdf
DESTINATION share/${PROJECT_NAME}/
)
ament_package())et package.xml (<?xml version="1.0"?>
<?xml-model href="http://download.ros.org/schema/package_format3.xsd" schematypens="http://www.w3.org/2001/XMLSchema"?>
<package format="3">
<name>tugbot_description</name>
<version>0.0.0</version>
<description>TODO: Package description</description>
<maintainer email="porizou.electro@gmail.com">porizou</maintainer>
<license>TODO: License declaration</license>
<buildtool_depend>ament_cmake</buildtool_depend>
<depend>robot_state_publisher</depend>
<depend>xacro</depend>
<test_depend>ament_lint_auto</test_depend>
```

<test_depend>ament_lint_common</test_depend>

<export>

<build_type>ament_cmake</build_type>

</export>

</package>)maintenant dans tugbot_gazebo il y a launch dans lequel il y a ign_ros2_bridge.laucnh.py(from launch import LaunchDescription

from launch.actions import DeclareLaunchArgument

from launch.substitutions import LaunchConfiguration

from launch_ros.actions import Node

def generate_launch_description():

return LaunchDescription([

DeclareLaunchArgument('use_sim_time', default_value='true',

description='Use simulation (Gazebo) clock if true'),

Node(

package='ros_ign_bridge',

executable='parameter_bridge',

name='bridge_node',

arguments=[

'/model/tugbot/cmd_vel@geometry_msgs/msg/Twist@ignition.msgs.Twist',

'/world/world_demo/model/tugbot/link/scan_omni/sensor/scan_omni/scan/points@sensor_msgs/msg/PointCloud2@ignition.msgs.PointCloudPacked',

'/world/world_demo/model/tugbot/link/camera_front/sensor/color/image@sensor_msgs/msg/Image@ignition.msgs.Image',

'/model/tugbot/tf@tf2_msgs/msg/TFMessage@ignition.msgs.Pose_V',

'/model/tugbot/odometry@nav_msgs/msg/Odometry@ignition.msgs.Odometry',

'/world/world_demo/model/tugbot/link/imu_link/sensor/imu/imu@sensor_msgs/msg/Imu@ignition.msgs.IMU',

'/clock@rosgraph_msgs/msg/Clock@ignition.msgs.Clock',

'/world/world_demo/model/tugbot/link/scan_front/sensor/scan_front/scan@sensor_msgs/msg/LaserScan@ignition.msgs.LaserScan',

'/world/world_demo/model/tugbot/link/scan_back/sensor/scan_back/scan@sensor_msgs/msg/LaserScan@ignition.msgs.LaserScan'

],

```
remappings=[
('/model/tugbot/tf', '/tf'),
('/model/tugbot/odometry', '/odom'),
],
output='screen',
parameters=[
{"use_sim_time": LaunchConfiguration('use_sim_time')}
],
),
])) et tugbot_depot.launch.py(from launch import LaunchDescription
from launch.actions import IncludeLaunchDescription
from launch.launch_description_sources import PythonLaunchDescriptionSource
from launch.substitutions import PathJoinSubstitution
from ament_index_python.packages import get_package_share_directory
import os
def generate_launch_description():
world_file_path = PathJoinSubstitution([
get_package_share_directory('tugbot_gazebo'),
'worlds',
'tugbot_depot.sdf'
])
ign_ros_bridge_launch_path = PathJoinSubstitution([
get_package_share_directory('tugbot_gazebo'),
'launch',
'ign_ros2_bridge.launch.py'
])
robot_description_launch_path = PathJoinSubstitution([
get_package_share_directory('tugbot_description'),
'launch',
```

'tugbot_description.launch.py'

])

return LaunchDescription([

IncludeLaunchDescription(

PythonLaunchDescriptionSource([os.path.join(

get_package_share_directory('ros_gz_sim'), 'launch'), '/gz_sim.launch.py']),

launch_arguments=[

('gz_args', world_file_path)]

),

IncludeLaunchDescription(

PythonLaunchDescriptionSource(ign_ros_bridge_launch_path)

),

IncludeLaunchDescription(

PythonLaunchDescriptionSource(robot_description_launch_path)

)

]))on sort de launch en parallele il y a worlds dedans il y a tugbot_depot.sdf(<?xml version='1.0' encoding='ASCII'?>

<sdf version='1.7'>

<world name='world_demo'>

<physics name='1ms' type='ignored'>

<max_step_size>0.01</max_step_size>

<real_time_factor>1</real_time_factor>

</physics>

<plugin name='ignition::gazebo::systems::Physics' filename='libignition-gazebo-physics-system.so'/>

<plugin name='ignition::gazebo::systems::UserCommands' filename='libignition-gazebo-user-commands-system.so'/>

<plugin name='ignition::gazebo::systems::SceneBroadcaster' filename='libignition-gazebo-scene-broadcaster-system.so'/>

<plugin name='ignition::gazebo::systems::Imu' filename='ignition-gazebo-imu-system'/>

<plugin name='ignition::gazebo::systems::Sensors' filename='ignition-gazebo-sensors-system'>

<render_engine>ogre2</render_engine>

```xml
</plugin>

<scene>

<ambient>1 1 1 1</ambient>

<background>0.3 0.7 0.9 1</background>

<shadows>0</shadows>

<grid>1</grid>

</scene>

<model name='ground_plane'>

<static>1</static>

<link name='link'>

<collision name='collision'>

<geometry>

<plane>

<normal>0 0 1</normal>

<size>1 1</size>

</plane>

</geometry>

<surface>

<friction>

<ode/>

</friction>

<bounce/>

<contact/>

</surface>

</collision>

</link>

<pose>0 0 0 0 0 0</pose>

</model>

<include>
```

```
<uri>
https://fuel.ignitionrobotics.org/1.0/OpenRobotics/models/Depot
</uri>
<pose>7.19009 1.09982 0 0 0 0</pose>
</include>
<model name='depot_collision'>
<static>1</static>
<pose>7.19009 1.09982 0 0 0 0</pose>
<link name='collision_link'>
<pose>0 0 0 0 0 0</pose>
<collision name='wall1'>
<pose>0 -7.6129 4.5 0 0 0</pose>
<geometry>
<box>
<size>30.167 0.08 9</size>
</box>
</geometry>
</collision>
<collision name='wall2'>
<pose>0 7.2875 4.5 0 0 0</pose>
<geometry>
<box>
<size>30.167 0.08 9</size>
</box>
</geometry>
</collision>
<collision name='wall3'>
<pose>-15 0 4.5 0 0 0</pose>
<geometry>
```

```
<box>
<size>0.08 15.360 9</size>
</box>
</geometry>
</collision>
<collision name='wall4'>
<pose>15 0 4.5 0 0 0</pose>
<geometry>
<box>
<size>0.08 15.360 9</size>
</box>
</geometry>
</collision>
<collision name='boxes1'>
<pose>0.22268 -4.7268 0.68506 0 0 0</pose>
<geometry>
<box>
<size>1.288 1.422 1.288</size>
</box>
</geometry>
</collision>
<collision name='boxes2'>
<pose>3.1727 -4.7268 0.68506 0 0 0</pose>
<geometry>
<box>
<size>1.288 1.422 1.288</size>
</box>
</geometry>
</collision>
```

```
<collision name='boxes3'>
<pose>5.95268 -4.7268 0.68506 0 0 0</pose>
<geometry>
<box>
<size>1.288 1.422 1.288</size>
</box>
</geometry>
</collision>
<collision name='boxes4'>
<pose>8.55887 -4.7268 0.68506 0 0 0</pose>
<geometry>
<box>
<size>1.288 1.422 1.288</size>
</box>
</geometry>
</collision>
<collision name='boxes5'>
<pose>11.326 -4.7268 0.68506 0 0 0</pose>
<geometry>
<box>
<size>1.288 1.422 1.288</size>
</box>
</geometry>
</collision>
<collision name='boxes6'>
<pose>0.22268 -2.37448 0.68506 0 0 0</pose>
<geometry>
<box>
<size>1.288 1.422 1.288</size>
```

</box>

</geometry>

</collision>

<collision name='boxes7'>

<pose>3.1727 -2.37448 0.68506 0 0 0</pose>

<geometry>

<box>

<size>1.288 1.422 1.288</size>

</box>

</geometry>

</collision>

<collision name='boxes8'>

<pose>5.95268 -2.37448 0.68506 0 0 0</pose>

<geometry>

<box>

<size>1.288 1.422 1.288</size>

</box>

</geometry>

</collision>

<collision name='boxes9'>

<pose>8.55887 -2.37448 0.68506 0 0 0</pose>

<geometry>

<box>

<size>1.288 1.422 1.288</size>

</box>

</geometry>

</collision>

<collision name='boxes10'>

<pose>11.326 -2.37448 0.68506 0 0 0</pose>

```
<geometry>
<box>
<size>1.288 1.422 1.288</size>
</box>
</geometry>
</collision>
<collision name='boxes11'>
<pose>-1.2268 4.1557 0.68506 0 0 -1.02799893</pose>
<geometry>
<box>
<size>1.288 1.422 1.288</size>
</box>
</geometry>
</collision>
<collision name='pilar1'>
<pose>-7.5402 3.6151 1 0 0 0</pose>
<geometry>
<box>
<size>0.465 0.465 2</size>
</box>
</geometry>
</collision>
<collision name='pilar2'>
<pose>7.4575 3.6151 1 0 0 0</pose>
<geometry>
<box>
<size>0.465 0.465 2</size>
</box>
</geometry>
```

</collision>

<collision name='pilar3'>

<pose>-7.5402 -3.8857 1 0 0 0</pose>

<geometry>

<box>

<size>0.465 0.465 2</size>

</box>

</geometry>

</collision>

<collision name='pilar4'>

<pose>7.4575 -3.8857 1 0 0 0</pose>

<geometry>

<box>

<size>0.465 0.465 2</size>

</box>

</geometry>

</collision>

<collision name='pallet_mover1'>

<pose>-0.6144 -2.389 0.41838 0 0 0</pose>

<geometry>

<box>

<size>0.363 0.440 0.719</size>

</box>

</geometry>

</collision>

<collision name='pallet_mover2'>

<pose>-1.6004 4.8225 0.41838 0 0 0</pose>

<geometry>

<box>

<size>0.363 0.244 0.719</size>

</box>

</geometry>

</collision>

<collision name='stairs'>

<pose>13.018 3.1652 0.25 0 0 0</pose>

<geometry>

<box>

<size>1.299 0.6 0.5</size>

</box>

</geometry>

</collision>

<collision name='shelfs1'>

<pose>1.4662 -0.017559 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs2'>

<pose>2.6483 -0.017559 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs3'>

<pose>5.3247 -0.017559 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs4'>

<pose>6.5063 -0.017559 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs5'>

<pose>9.0758 -0.017559 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs6'>

<pose>10.258 -0.017559 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs7'>

<pose>1.4662 2.5664 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs8'>

<pose>2.6483 2.5664 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs9'>

<pose>5.3247 2.5664 0.5 0 0 0</pose>

<geometry>

```
<cylinder>
<radius>0.03</radius>
<length>1</length>
</cylinder>
</geometry>
</collision>
<collision name='shelfs10'>
<pose>6.5063 2.5664 0.5 0 0 0</pose>
<geometry>
<cylinder>
<radius>0.03</radius>
<length>1</length>
</cylinder>
</geometry>
</collision>
<collision name='shelfs11'>
<pose>9.0758 2.5664 0.5 0 0 0</pose>
<geometry>
<cylinder>
<radius>0.03</radius>
<length>1</length>
</cylinder>
</geometry>
</collision>
<collision name='shelfs12'>
<pose>10.258 2.5664 0.5 0 0 0</pose>
<geometry>
<cylinder>
<radius>0.03</radius>
</cylinder>
```

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs13'>

<pose>1.4662 5.1497 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs14'>

<pose>2.6483 5.1497 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs15'>

<pose>5.3247 5.1497 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs16'>

<pose>6.5063 5.1497 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs17'>

<pose>9.0758 5.1497 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

<collision name='shelfs18'>

<pose>10.258 5.1497 0.5 0 0 0</pose>

<geometry>

<cylinder>

<radius>0.03</radius>

<length>1</length>

</cylinder>

</geometry>

</collision>

</link>

</model>

<include>

<uri>

https://fuel.ignitionrobotics.org/1.0/MovAi/models/Tugbot

</uri>

<name>tugbot</name>

<pose>2.75268 1.0014 0.132279 0 0 0</pose>

</include>

</world>

</sdf>)il y a aussi dans tugbot_gazebo CMakeLists.txt (cmake_minimum_required(VERSION 3.8)

project(tugbot_gazebo)

if(CMAKE_COMPILER_IS_GNUCXX OR CMAKE_CXX_COMPILER_ID MATCHES "Clang")

add_compile_options(-Wall -Wextra -Wpedantic)

endif()

find dependencies

find_package(ament_cmake REQUIRED)

uncomment the following section in order to fill in

further dependencies manually.

find_package(<dependency> REQUIRED)

if(BUILD_TESTING)

find_package(ament_lint_auto REQUIRED)

the following line skips the linter which checks for copyrights

comment the line when a copyright and license is added to all source files

set(ament_cmake_copyright_FOUND TRUE)

the following line skips cpplint (only works in a git repo)

comment the line when this package is in a git repo and when

a copyright and license is added to all source files

set(ament_cmake_cpplint_FOUND TRUE)

ament_lint_auto_find_test_dependencies()

endif()

install(DIRECTORY

launch

worlds

DESTINATION share/${PROJECT_NAME}/

)

ament_package())et package.xml (<?xml version="1.0"?>

<?xml-model href="http://download.ros.org/schema/package_format3.xsd" schematypens="http://www.w3.org/2001/XMLSchema"?>

<package format="3">

<name>tugbot_gazebo</name>

<version>0.0.0</version>

<description>TODO: Package description</description>

<maintainer email="porizou.electro@gmail.com">porizou</maintainer>

<license>TODO: License declaration</license>

<buildtool_depend>ament_cmake</buildtool_depend>

<test_depend>ament_lint_auto</test_depend>

<test_depend>ament_lint_common</test_depend>

<export>

<build_type>ament_cmake</build_type>

</export>

</package>) on sort de tugbot_gazebo totalement on va vers tugbot_navigation2 il ya dedans config ou dedans il y a navigation2.yaml (amcl:

ros__parameters:

use_sim_time: False

alpha1: 0.2

alpha2: 0.2

alpha3: 0.2

alpha4: 0.2

alpha5: 0.2

base_frame_id: "base_link"

beam_skip_distance: 0.5

beam_skip_error_threshold: 0.9

beam_skip_threshold: 0.3

do_beamskip: false

global_frame_id: "map"

lambda_short: 0.1

laser_likelihood_max_dist: 2.0

laser_max_range: 100.0

laser_min_range: -1.0

laser_model_type: "likelihood_field"

max_beams: 60

max_particles: 2000

min_particles: 500

odom_frame_id: "odom"

pf_err: 0.05

pf_z: 0.99

recovery_alpha_fast: 0.0

recovery_alpha_slow: 0.0

resample_interval: 1

robot_model_type: "nav2_amcl::DifferentialMotionModel"

save_pose_rate: 0.5

sigma_hit: 0.2

tf_broadcast: true

transform_tolerance: 1.0

update_min_a: 0.2

update_min_d: 0.25

z_hit: 0.5

z_max: 0.05

z_rand: 0.5

z_short: 0.05

scan_topic: scan

amcl_map_client:

ros__parameters:

use_sim_time: False

amcl_rclcpp_node:

ros__parameters:

use_sim_time: False

bt_navigator:

ros__parameters:

use_sim_time: False

global_frame: map

robot_base_frame: base_link

odom_topic: /odom

default_bt_xml_filename: "navigate_w_replanning_and_recovery.xml"

bt_loop_duration: 10

default_server_timeout: 20

enable_groot_monitoring: True

groot_zmq_publisher_port: 1666

groot_zmq_server_port: 1667

plugin_lib_names:

- nav2_compute_path_to_pose_action_bt_node

- nav2_compute_path_through_poses_action_bt_node

- nav2_follow_path_action_bt_node

- nav2_back_up_action_bt_node

- nav2_spin_action_bt_node

- nav2_wait_action_bt_node

- nav2_clear_costmap_service_bt_node

- nav2_is_stuck_condition_bt_node

- nav2_goal_reached_condition_bt_node

- nav2_goal_updated_condition_bt_node

- nav2_initial_pose_received_condition_bt_node

- nav2_reinitialize_global_localization_service_bt_node

- nav2_rate_controller_bt_node

- nav2_distance_controller_bt_node

- nav2_speed_controller_bt_node

- nav2_truncate_path_action_bt_node

- nav2_goal_updater_node_bt_node

- nav2_recovery_node_bt_node

- nav2_pipeline_sequence_bt_node

- nav2_round_robin_node_bt_node

- nav2_transform_available_condition_bt_node

- nav2_time_expired_condition_bt_node

- nav2_distance_traveled_condition_bt_node

- nav2_single_trigger_bt_node

- nav2_is_battery_low_condition_bt_node

- nav2_navigate_through_poses_action_bt_node

- nav2_navigate_to_pose_action_bt_node

- nav2_remove_passed_goals_action_bt_node

- nav2_planner_selector_bt_node

- nav2_controller_selector_bt_node

- nav2_goal_checker_selector_bt_node

bt_navigator_rclcpp_node:

ros__parameters:

use_sim_time: False

controller_server:

ros__parameters:

use_sim_time: False

controller_frequency: 10.0

min_x_velocity_threshold: 0.001

min_y_velocity_threshold: 0.5

min_theta_velocity_threshold: 0.001

failure_tolerance: 0.3

progress_checker_plugin: "progress_checker"

goal_checker_plugins: ["general_goal_checker"]

controller_plugins: ["FollowPath"]

Progress checker parameters

progress_checker:

plugin: "nav2_controller::SimpleProgressChecker"

required_movement_radius: 0.5

movement_time_allowance: 10.0

general_goal_checker:

stateful: True

plugin: "nav2_controller::SimpleGoalChecker"

xy_goal_tolerance: 0.25

yaw_goal_tolerance: 0.25

DWB parameters

FollowPath:

plugin: "dwb_core::DWBLocalPlanner"

debug_trajectory_details: True

min_vel_x: 0.0

min_vel_y: 0.0

max_vel_x: 0.22

max_vel_y: 0.0

max_vel_theta: 1.0

min_speed_xy: 0.0

max_speed_xy: 0.22

min_speed_theta: 0.0

Add high threshold velocity for turtlebot 3 issue.

https://github.com/ROBOTIS-GIT/turtlebot3_simulations/issues/75

acc_lim_x: 2.5

acc_lim_y: 0.0

acc_lim_theta: 3.2

decel_lim_x: -2.5

decel_lim_y: 0.0

decel_lim_theta: -3.2

vx_samples: 20

vy_samples: 0

vtheta_samples: 40

sim_time: 1.5

linear_granularity: 0.05

angular_granularity: 0.025

transform_tolerance: 0.2

xy_goal_tolerance: 0.05

trans_stopped_velocity: 0.25

short_circuit_trajectory_evaluation: True

stateful: True

critics: ["RotateToGoal", "Oscillation", "BaseObstacle", "GoalAlign", "PathAlign", "PathDist", "GoalDist"]

BaseObstacle.scale: 0.02

PathAlign.scale: 32.0

PathAlign.forward_point_distance: 0.1

GoalAlign.scale: 24.0

GoalAlign.forward_point_distance: 0.1

PathDist.scale: 32.0

GoalDist.scale: 24.0

RotateToGoal.scale: 32.0

RotateToGoal.slowing_factor: 5.0

RotateToGoal.lookahead_time: -1.0

controller_server_rclcpp_node:

ros__parameters:

use_sim_time: False

local_costmap:

local_costmap:

ros__parameters:

update_frequency: 5.0

publish_frequency: 2.0

global_frame: odom

robot_base_frame: base_link

use_sim_time: False

rolling_window: true

width: 3

height: 3

resolution: 0.05

robot_radius: 0.1

plugins: ["obstacle_layer", "voxel_layer", "inflation_layer"]

inflation_layer:

plugin: "nav2_costmap_2d::InflationLayer"

inflation_radius: 1.0

cost_scaling_factor: 3.0

obstacle_layer:

plugin: "nav2_costmap_2d::ObstacleLayer"

enabled: True

observation_sources: scan

scan:

topic: /scan

max_obstacle_height: 2.0

clearing: True

marking: True

data_type: "LaserScan"

voxel_layer:

plugin: "nav2_costmap_2d::VoxelLayer"

enabled: True

publish_voxel_map: True

origin_z: 0.0

z_resolution: 0.05

z_voxels: 16

max_obstacle_height: 2.0

mark_threshold: 0

observation_sources: scan

scan:

topic: /scan

max_obstacle_height: 2.0

clearing: True

marking: True

data_type: "LaserScan"

raytrace_max_range: 3.0

raytrace_min_range: 0.0

obstacle_max_range: 2.5

obstacle_min_range: 0.0

static_layer:

map_subscribe_transient_local: True

always_send_full_costmap: True

```
local_costmap_client:
ros__parameters:
use_sim_time: False
local_costmap_rclcpp_node:
ros__parameters:
use_sim_time: False
global_costmap:
global_costmap:
ros__parameters:
update_frequency: 1.0
publish_frequency: 1.0
global_frame: map
robot_base_frame: base_link
use_sim_time: True
robot_radius: 0.1
resolution: 0.05
track_unknown_space: true
plugins: ["static_layer", "obstacle_layer", "voxel_layer", "inflation_layer"]
obstacle_layer:
plugin: "nav2_costmap_2d::ObstacleLayer"
enabled: True
observation_sources: scan
scan:
topic: /scan
max_obstacle_height: 2.0
clearing: True
marking: True
data_type: "LaserScan"
raytrace_max_range: 3.0
```

raytrace_min_range: 0.0

obstacle_max_range: 2.5

obstacle_min_range: 0.0

voxel_layer:

plugin: "nav2_costmap_2d::VoxelLayer"

enabled: True

publish_voxel_map: True

origin_z: 0.0

z_resolution: 0.05

z_voxels: 16

max_obstacle_height: 2.0

mark_threshold: 0

observation_sources: scan

scan:

topic: /scan

max_obstacle_height: 2.0

clearing: True

marking: True

data_type: "LaserScan"

raytrace_max_range: 3.0

raytrace_min_range: 0.0

obstacle_max_range: 2.5

obstacle_min_range: 0.0

static_layer:

plugin: "nav2_costmap_2d::StaticLayer"

map_subscribe_transient_local: True

inflation_layer:

plugin: "nav2_costmap_2d::InflationLayer"

cost_scaling_factor: 3.0

inflation_radius: 0.55

always_send_full_costmap: True

global_costmap_client:

ros__parameters:

use_sim_time: False

global_costmap_rclcpp_node:

ros__parameters:

use_sim_time: False

map_server:

ros__parameters:

use_sim_time: False

yaml_filename: "map.yaml"

map_saver:

ros__parameters:

use_sim_time: False

save_map_timeout: 5.0

free_thresh_default: 0.25

occupied_thresh_default: 0.65

map_subscribe_transient_local: True

planner_server:

ros__parameters:

expected_planner_frequency: 20.0

use_sim_time: False

planner_plugins: ["GridBased"]

GridBased:

plugin: "nav2_navfn_planner/NavfnPlanner"

tolerance: 0.5

use_astar: false

allow_unknown: true

planner_server_rclcpp_node:

ros__parameters:

use_sim_time: False

recoveries_server:

ros__parameters:

costmap_topic: local_costmap/costmap_raw

footprint_topic: local_costmap/published_footprint

cycle_frequency: 10.0

recovery_plugins: ["spin", "backup", "wait"]

spin:

plugin: "nav2_recoveries/Spin"

backup:

plugin: "nav2_recoveries/BackUp"

wait:

plugin: "nav2_recoveries/Wait"

global_frame: odom

robot_base_frame: base_link

transform_timeout: 0.1

use_sim_time: true

simulate_ahead_time: 2.0

max_rotational_vel: 1.0

min_rotational_vel: 0.4

rotational_acc_lim: 3.2

robot_state_publisher:

ros__parameters:

use_sim_time: False

waypoint_follower:

ros__parameters:

loop_rate: 2000

stop_on_failure: false

waypoint_task_executor_plugin: "wait_at_waypoint"

wait_at_waypoint:

plugin: "nav2_waypoint_follower::WaitAtWaypoint"

enabled: True

waypoint_pause_duration: 200)dans tugbot_navigation2 il y a launch dans lequel il y a navigation2.launch.py (import os

from ament_index_python.packages import get_package_share_directory

from launch import LaunchDescription

from launch.actions import DeclareLaunchArgument

from launch.actions import IncludeLaunchDescription

from launch.launch_description_sources import PythonLaunchDescriptionSource

from launch.substitutions import LaunchConfiguration

from launch_ros.actions import Node

def generate_launch_description():

use_sim_time = LaunchConfiguration('use_sim_time', default='true')

map_dir = LaunchConfiguration(

'map',

default=os.path.join(

get_package_share_directory('tugbot_navigation2'),

'map',

'map.yaml'))

param_file_name = 'navigation2.yaml'

param_dir = LaunchConfiguration(

'params_file',

default=os.path.join(

get_package_share_directory('tugbot_navigation2'),

'config',

param_file_name))

nav2_launch_file_dir = os.path.join(get_package_share_directory('nav2_bringup'), 'launch')

```
rviz_config_dir = os.path.join(

get_package_share_directory('tugbot_navigation2'),

'rviz',

'nav2.rviz')

return LaunchDescription([

DeclareLaunchArgument(

'map',

default_value=map_dir,

description='Full path to map file to load'),

DeclareLaunchArgument(

'params_file',

default_value=param_dir,

description='Full path to param file to load'),

DeclareLaunchArgument(

'use_sim_time',

default_value='false',

description='Use simulation (Gazebo) clock if true'),

IncludeLaunchDescription(

PythonLaunchDescriptionSource([nav2_launch_file_dir, '/bringup_launch.py']),

launch_arguments={

'map': map_dir,

'use_sim_time': use_sim_time,

'params_file': param_dir}.items(),

),

Node(

package='rviz2',

executable='rviz2',

name='rviz2',

arguments=['-d', rviz_config_dir],
```

parameters=[{'use_sim_time': use_sim_time}],

output='screen'),

]))ensuite dans rviz il y a nav2.rviz (Panels:

- Class: rviz_common/Displays

Help Height: 0

Name: Displays

Property Tree Widget:

Expanded:

- /Global Options1

- /TF1

- /TF1/Frames1

- /TF1/Tree1

Splitter Ratio: 0.5833333134651184

Tree Height: 462

- Class: rviz_common/Selection

Name: Selection

- Class: rviz_common/Tool Properties

Expanded:

- /Publish Point1

Name: Tool Properties

Splitter Ratio: 0.5886790156364441

- Class: rviz_common/Views

Expanded:

- /Current View1

Name: Views

Splitter Ratio: 0.5

- Class: nav2_rviz_plugins/Navigation 2

Name: Navigation 2

Visualization Manager:

Class: ""

Displays:

- Alpha: 0.5

Cell Size: 1

Class: rviz_default_plugins/Grid

Color: 160; 160; 164

Enabled: true

Line Style:

Line Width: 0.029999999329447746

Value: Lines

Name: Grid

Normal Cell Count: 0

Offset:

X: 0

Y: 0

Z: 0

Plane: XY

Plane Cell Count: 10

Reference Frame: <Fixed Frame>

Value: true

- Alpha: 1

Class: rviz_default_plugins/RobotModel

Collision Enabled: false

Description File: ""

Description Source: Topic

Description Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /robot_description

Enabled: true

Links:

All Links Enabled: true

Expand Joint Details: false

Expand Link Details: false

Expand Tree: false

Link Tree Style: ""

Mass Properties:

Inertia: false

Mass: false

Name: RobotModel

TF Prefix: ""

Update Interval: 0

Value: true

Visual Enabled: true

- Class: rviz_default_plugins/TF

Enabled: true

Frame Timeout: 15

Frames:

All Enabled: false

Marker Scale: 1

Name: TF

Show Arrows: true

Show Axes: true

Show Names: true

Tree:

{}

Update Interval: 0

Value: true

- Alpha: 1

Autocompute Intensity Bounds: true

Autocompute Value Bounds:

Max Value: 10

Min Value: -10

Value: true

Axis: Z

Channel Name: intensity

Class: rviz_default_plugins/LaserScan

Color: 255; 255; 255

Color Transformer: Intensity

Decay Time: 0

Enabled: true

Invert Rainbow: false

Max Color: 255; 255; 255

Max Intensity: 0

Min Color: 0; 0; 0

Min Intensity: 0

Name: LaserScan

Position Transformer: XYZ

Selectable: true

Size (Pixels): 3

Size (m): 0.009999999776482582

Style: Points

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Best Effort

Value: /scan

Use Fixed Frame: true

Use rainbow: true

Value: true

- Alpha: 1

Autocompute Intensity Bounds: true

Autocompute Value Bounds:

Max Value: 10

Min Value: -10

Value: true

Axis: Z

Channel Name: intensity

Class: rviz_default_plugins/PointCloud2

Color: 255; 255; 255

Color Transformer: ""

Decay Time: 0

Enabled: true

Invert Rainbow: false

Max Color: 255; 255; 255

Max Intensity: 4096

Min Color: 0; 0; 0

Min Intensity: 0

Name: Bumper Hit

Position Transformer: ""

Selectable: true

Size (Pixels): 3

Size (m): 0.07999999821186066

Style: Spheres

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Best Effort

Value: /mobile_base/sensors/bumper_pointcloud

Use Fixed Frame: true

Use rainbow: true

Value: true

- Alpha: 1

Class: rviz_default_plugins/Map

Color Scheme: map

Draw Behind: true

Enabled: true

Name: Map

Topic:

Depth: 1

Durability Policy: Transient Local

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /map

Update Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /map_updates

Use Timestamp: false

Value: true

- Alpha: 1

Class: nav2_rviz_plugins/ParticleCloud

Color: 0; 180; 0

Enabled: true

Max Arrow Length: 0.30000001192092896

Min Arrow Length: 0.019999999552965164

Name: Amcl Particle Swarm

Shape: Arrow (Flat)

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Best Effort

Value: /particle_cloud

Value: true

- Class: rviz_common/Group

Displays:

- Alpha: 0.30000001192092896

Class: rviz_default_plugins/Map

Color Scheme: costmap

Draw Behind: false

Enabled: true

Name: Global Costmap

Topic:

Depth: 1

Durability Policy: Transient Local

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /global_costmap/costmap

Update Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /global_costmap/costmap_updates

Use Timestamp: false

Value: true

- Alpha: 0.30000001192092896

Class: rviz_default_plugins/Map

Color Scheme: costmap

Draw Behind: false

Enabled: true

Name: Downsampled Costmap

Topic:

Depth: 1

Durability Policy: Transient Local

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /downsampled_costmap

Update Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /downsampled_costmap_updates

Use Timestamp: false

Value: true

- Alpha: 1

Buffer Length: 1

Class: rviz_default_plugins/Path

Color: 255; 0; 0

Enabled: true

Head Diameter: 0.019999999552965164

Head Length: 0.019999999552965164

Length: 0.30000001192092896

Line Style: Lines

Line Width: 0.029999999329447746

Name: Path

Offset:

X: 0

Y: 0

Z: 0

Pose Color: 255; 85; 255

Pose Style: Arrows

Radius: 0.029999999329447746

Shaft Diameter: 0.004999999888241291

Shaft Length: 0.019999999552965164

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /plan

Value: true

- Alpha: 1

Autocompute Intensity Bounds: true

Autocompute Value Bounds:

Max Value: 10

Min Value: -10

Value: true

Axis: Z

Channel Name: intensity

Class: rviz_default_plugins/PointCloud2

Color: 125; 125; 125

Color Transformer: FlatColor

Decay Time: 0

Enabled: true

Invert Rainbow: false

Max Color: 255; 255; 255

Max Intensity: 4096

Min Color: 0; 0; 0

Min Intensity: 0

Name: VoxelGrid

Position Transformer: XYZ

Selectable: true

Size (Pixels): 3

Size (m): 0.05000000074505806

Style: Boxes

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /global_costmap/voxel_marked_cloud

Use Fixed Frame: true

Use rainbow: true

Value: true

- Alpha: 1

Class: rviz_default_plugins/Polygon

Color: 25; 255; 0

Enabled: false

Name: Polygon

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /global_costmap/published_footprint

Value: false

Enabled: true

Name: Global Planner

- Class: rviz_common/Group

Displays:

- Alpha: 0.699999988079071

Class: rviz_default_plugins/Map

Color Scheme: costmap

Draw Behind: false

Enabled: true

Name: Local Costmap

Topic:

Depth: 1

Durability Policy: Transient Local

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /local_costmap/costmap

Update Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /local_costmap/costmap_updates

Use Timestamp: false

Value: true

- Alpha: 1

Buffer Length: 1

Class: rviz_default_plugins/Path

Color: 0; 12; 255

Enabled: true

Head Diameter: 0.30000001192092896

Head Length: 0.20000000298023224

Length: 0.30000001192092896

Line Style: Lines

Line Width: 0.029999999329447746

Name: Local Plan

Offset:

X: 0

Y: 0

Z: 0

Pose Color: 255; 85; 255

Pose Style: None

Radius: 0.029999999329447746

Shaft Diameter: 0.10000000149011612

Shaft Length: 0.10000000149011612

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /local_plan

Value: true

- Class: rviz_default_plugins/MarkerArray

Enabled: false

Name: Trajectories

Namespaces:

{}

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /marker

Value: false

- Alpha: 1

Class: rviz_default_plugins/Polygon

Color: 25; 255; 0

Enabled: true

Name: Polygon

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /local_costmap/published_footprint

Value: true

- Alpha: 1

Autocompute Intensity Bounds: true

Autocompute Value Bounds:

Max Value: 10

Min Value: -10

Value: true

Axis: Z

Channel Name: intensity

Class: rviz_default_plugins/PointCloud2

Color: 255; 255; 255

Color Transformer: RGB8

Decay Time: 0

Enabled: true

Invert Rainbow: false

Max Color: 255; 255; 255

Max Intensity: 4096

Min Color: 0; 0; 0

Min Intensity: 0

Name: VoxelGrid

Position Transformer: XYZ

Selectable: true

Size (Pixels): 3

Size (m): 0.009999999776482582

Style: Flat Squares

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /local_costmap/voxel_marked_cloud

Use Fixed Frame: true

Use rainbow: true

Value: true

Enabled: true

Name: Controller

- Class: rviz_common/Group

Displays:

- Class: rviz_default_plugins/Image

Enabled: true

Max Value: 1

Median window: 5

Min Value: 0

Name: RealsenseCamera

Normalize Range: true

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /intel_realsense_r200_depth/image_raw

Value: true

- Alpha: 1

Autocompute Intensity Bounds: true

Autocompute Value Bounds:

Max Value: 10

Min Value: -10

Value: true

Axis: Z

Channel Name: intensity

Class: rviz_default_plugins/PointCloud2

Color: 255; 255; 255

Color Transformer: RGB8

Decay Time: 0

Enabled: true

Invert Rainbow: false

Max Color: 255; 255; 255

Max Intensity: 4096

Min Color: 0; 0; 0

Min Intensity: 0

Name: RealsenseDepthImage

Position Transformer: XYZ

Selectable: true

Size (Pixels): 3

Size (m): 0.009999999776482582

Style: Flat Squares

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /intel_realsense_r200_depth/points

Use Fixed Frame: true

Use rainbow: true

Value: true

Enabled: false

Name: Realsense

- Class: rviz_default_plugins/MarkerArray

Enabled: true

Name: MarkerArray

Namespaces:

{}

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /waypoints

Value: true

Enabled: true

Global Options:

Background Color: 48; 48; 48

Fixed Frame: map

Frame Rate: 30

Name: root

Tools:

- Class: rviz_default_plugins/MoveCamera

- Class: rviz_default_plugins/Select

- Class: rviz_default_plugins/FocusCamera

- Class: rviz_default_plugins/Measure

Line color: 128; 128; 0

- Class: rviz_default_plugins/SetInitialPose

Covariance x: 0.25

Covariance y: 0.25

Covariance yaw: 0.06853891909122467

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /initialpose

- Class: rviz_default_plugins/PublishPoint

Single click: true

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /clicked_point

- Class: nav2_rviz_plugins/GoalTool

Transformation:

Current:

Class: rviz_default_plugins/TF

Value: true

Views:

Current:

Angle: -1.6150002479553223

Class: rviz_default_plugins/TopDownOrtho

Enable Stereo Rendering:

Stereo Eye Separation: 0.05999999865889549

Stereo Focal Distance: 1

Swap Stereo Eyes: false

Value: false

Invert Z Axis: false

Name: Current View

Near Clip Distance: 0.009999999776482582

Scale: 127.88431549072266

Target Frame: <Fixed Frame>

Value: TopDownOrtho (rviz_default_plugins)

X: -0.044467076659202576

Y: -0.38726311922073364

Saved: ~

Window Geometry:

Displays:

collapsed: false

Height: 932

Hide Left Dock: false

Hide Right Dock: true

Navigation 2:

collapsed: false

QMainWindow State:
000000ff00000000fd00000004000000000000016a0000034afc020000000afb0000001200530065006c00650063
00740069006f006e00000001e10000009b0000005c00fffffffb0000001e0054006f006f006c002000500072006f0070
006500720074006900650073020000001ed000001df00000185000000a3fb000000120056006900650077007300200054006f006f006f02000001df000002110000018500000122fb000000200054006f006f006c002000500072006f0070
0065007200740069006500730032030000028800000011d00000221000017afb000000100044006900730070006c006100790073010000003d0000020b000000c900fffffffb0000002000730065006c0065006300740069006f006e
00200062007500660066006500720200000138000000aa0000023a00000294fb00000014005700690064006500
530074006500720065006f02000000e6000000d2000003ee0000030bfb0000000c004b0069006e0065006300740
200000186000001060000030c00000261fb00000018004e0061007600690067006100740069006f006e00200032
010000024e000001390000013900fffffffb0000001e005200650061006c00730065006e0073006500430061006d00
6500720061000000002c6000000c10000002800ffffff000000010000010f0000034afc0200000003fb0000001e00540
06f006f006c002000500072006f00700065007200740069006500730100000041000000780000000000000000fb0
000000a005600690065007700730000000003d0000034a000000a400fffffffb0000001200530065006c00650063007
40069006f006e010000025a000000b200000000000000000000000200000490000000a9fc0100000001fb000000
0a00560069006500770073030000004e00000080000002e100000197000000030000044200000003efc01000000
02fb0000000800540069006d006501000000000000044200000000000000000fb0000000800540069006d0065010
000000000000450000000000000000000000004990000034a00000004000000040000000800000008fc000000010
00000020000000010000000a0054006f006f006c00730100000000ffffffff0000000000000000

RealsenseCamera:

collapsed: false

Selection:

collapsed: false

Tool Properties:

collapsed: false

Views:

collapsed: true

Width: 1545

X: 196

Y: 102), dans scripts il y a save_map.sh(#!/bin/bash

map_name="map"

if ["$1"]; then

map_name=$1

fi

map_saverを実行して地図を保存

ros2 run nav2_map_server map_saver -f ../map/$map_name), CMakeLists.txt
(cmake_minimum_required(VERSION 3.8)

project(tugbot_navigation2)

```
if(CMAKE_COMPILER_IS_GNUCXX OR CMAKE_CXX_COMPILER_ID MATCHES "Clang")

add_compile_options(-Wall -Wextra -Wpedantic)

endif()

# find dependencies

find_package(ament_cmake REQUIRED)

# uncomment the following section in order to fill in

# further dependencies manually.

# find_package(<dependency> REQUIRED)

if(BUILD_TESTING)

find_package(ament_lint_auto REQUIRED)

# the following line skips the linter which checks for copyrights

# comment the line when a copyright and license is added to all source files

set(ament_cmake_copyright_FOUND TRUE)

# the following line skips cpplint (only works in a git repo)

# comment the line when this package is in a git repo and when

# a copyright and license is added to all source files

set(ament_cmake_cpplint_FOUND TRUE)

ament_lint_auto_find_test_dependencies()

endif()

install(DIRECTORY

launch

config

rviz

map

scripts

DESTINATION share/${PROJECT_NAME}/

)

ament_package()),package.xml(<?xml version="1.0"?>

<?xml-model href="http://download.ros.org/schema/package_format3.xsd"
schematypens="http://www.w3.org/2001/XMLSchema"?>
```

<package format="3">

<name>tugbot_navigation2</name>

<version>0.0.0</version>

<description>TODO: Package description</description>

<maintainer email="porizou.electro@gmail.com">porizou</maintainer>

<license>TODO: License declaration</license>

<buildtool_depend>ament_cmake</buildtool_depend>

<test_depend>ament_lint_auto</test_depend>

<test_depend>ament_lint_common</test_depend>

<export>

<build_type>ament_cmake</build_type>

</export>

</package>)maintenant on sors de tugbot_navigation2 on va dans tugbot_ros2_pkgs il y a dedans CMakeLists.txt (cmake_minimum_required(VERSION 3.8)

project(tugbot_ros2_pkgs)

if(CMAKE_COMPILER_IS_GNUCXX OR CMAKE_CXX_COMPILER_ID MATCHES "Clang")

add_compile_options(-Wall -Wextra -Wpedantic)

endif()

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ament_lint_auto_find_test_dependencies()

endif()

ament_package())et package.xml(<?xml version="1.0"?>

<?xml-model href="http://download.ros.org/schema/package_format3.xsd" schematypens="http://www.w3.org/2001/XMLSchema"?>

<package format="3">

<name>tugbot_ros2_pkgs</name>

<version>0.0.0</version>

<description>TODO: Package description</description>

<maintainer email="porizou.electro@gmail.com">porizou</maintainer>

<license>TODO: License declaration</license>

<buildtool_depend>ament_cmake</buildtool_depend>

<test_depend>ament_lint_auto</test_depend>

<test_depend>ament_lint_common</test_depend>

<exec_depend>std_msgs</exec_depend>

<export>

<build_type>ament_cmake</build_type>

</export>

</package>)maintenant on va dans tugbot_ros2_pkgs il y a config dans laquelle il y a mapper_params_offline.yaml(# Ref: https://github.com/SteveMacenski/slam_toolbox/blob/dashing-devel/config/mapper_params_offline.yaml

slam_toolbox:

ros__parameters:

Plugin params

solver_plugin: solver_plugins::CeresSolver

ceres_linear_solver: SPARSE_NORMAL_CHOLESKY

ceres_preconditioner: SCHUR_JACOBI

ceres_trust_strategy: LEVENBERG_MARQUARDT

ceres_dogleg_type: TRADITIONAL_DOGLEG

```yaml
ceres_loss_function: None

# ROS Parameters

odom_frame: odom

map_frame: map

base_frame: base_link

scan_topic: /scan

mode: mapping #localization

debug_logging: false

throttle_scans: 1

transform_publish_period: 0.02 #if 0 never publishes odometry

map_update_interval: 10.0

resolution: 0.05

max_laser_range: 20.0 #for rastering images

minimum_time_interval: 0.5

transform_timeout: 0.2

tf_buffer_duration: 14400.

stack_size_to_use: 40000000 #// program needs a larger stack size to serialize large maps

enable_interactive_mode: true

use_sim_time: true

# General Parameters

use_scan_matching: true

use_scan_barycenter: true

minimum_travel_distance: 0.5

minimum_travel_heading: 0.5

scan_buffer_size: 10

scan_buffer_maximum_scan_distance: 10.0

link_match_minimum_response_fine: 0.1

link_scan_maximum_distance: 1.5

loop_search_maximum_distance: 3.0
```

do_loop_closing: true

loop_match_minimum_chain_size: 10

loop_match_maximum_variance_coarse: 3.0

loop_match_minimum_response_coarse: 0.35

loop_match_minimum_response_fine: 0.45

Correlation Parameters - Correlation Parameters

correlation_search_space_dimension: 0.5

correlation_search_space_resolution: 0.01

correlation_search_space_smear_deviation: 0.1

Correlation Parameters - Loop Closure Parameters

loop_search_space_dimension: 8.0

loop_search_space_resolution: 0.05

loop_search_space_smear_deviation: 0.03

Scan Matcher Parameters

distance_variance_penalty: 0.5

angle_variance_penalty: 1.0

fine_search_angle_offset: 0.00349

coarse_search_angle_offset: 0.349

coarse_angle_resolution: 0.0349

minimum_angle_penalty: 0.9

minimum_distance_penalty: 0.5

use_response_expansion: true), launch dans lequel il y a slam_toolbox.launch.py (from ament_index_python.packages import get_package_share_directory

from launch import LaunchDescription

from launch_ros.actions import Node

from launch.actions import LogInfo, ExecuteProcess

def generate_launch_description():

slam_node = Node(

package='slam_toolbox', executable='sync_slam_toolbox_node',

output='screen',

```
parameters=[
get_package_share_directory(
'tugbot_slam')
+ '/config/mapper_params_offline.yaml'
],
remappings=[
('/scan', '/scan'),
],
)
rviz2_node = Node(
name='rviz2',
package='rviz2', executable='rviz2', output='screen',
arguments=[
'-d',
get_package_share_directory('tugbot_slam')
+ '/rviz/default.rviz'],
)
pointcloud_to_laserscan = Node(
package='pointcloud_to_laserscan',
executable='pointcloud_to_laserscan_node',
name='pointcloud_to_laserscan',
output='screen',
parameters=[
{'target_frame': 'base_link'}, # ターゲットフレーム
{'transform_tolerance': 0.01}, # 変換の許容誤差
{'min_height': 0.5}, # 最小高さ
{'max_height': 1.5}, # 最大高さ
{'angle_min': -1.57}, # 最小角度（ラジアン）
{'angle_max': 1.57}, # 最大角度（ラジアン）
```

{'angle_increment': 0.01}, # 角度の増分（ラジアン）

{'scan_time': 0.1}, # スキャン時間（秒）

{'range_min': 0.3}, # 最小範囲（メートル）

{'range_max': 30.0}, # 最大範囲（メートル）

{'use_inf': True}, # 無限遠を使用するかどうか

],

remappings=[

('cloud_in', '/world/world_demo/model/tugbot/link/scan_omni/sensor/scan_omni/scan/points'), # 入力PointCloud2 トピック

('scan', 'scan') # 出力LaserScanトピック

]

)

teleop_twist_keyboard_node = ExecuteProcess(

cmd=['gnome-terminal', '--', 'ros2', 'run', 'teleop_twist_keyboard', 'teleop_twist_keyboard'],

shell=False,

output='screen'

)

ld = LaunchDescription()

ld.add_action(slam_node)

ld.add_action(rviz2_node)

ld.add_action(pointcloud_to_laserscan)

ld.add_action(LogInfo(msg="Opening a new terminal window for teleop_twist_keyboard..."))

ld.add_action(teleop_twist_keyboard_node)

return ld), rviz dans lequel il y a default.rviz (Panels:

- Class: rviz_common/Displays

Help Height: 78

Name: Displays

Property Tree Widget:

Expanded:

- /Global Options1

- /Status1

- /Odometry1/Shape1

- /LaserScan1

- /LaserScan1/Topic1

- /TF1/Frames1

- /RobotModel1

- /PointCloud21

Splitter Ratio: 0.5

Tree Height: 907

- Class: rviz_common/Selection

Name: Selection

- Class: rviz_common/Tool Properties

Expanded:

- /Publish Point1

Name: Tool Properties

Splitter Ratio: 0.5886790156364441

- Class: rviz_common/Views

Expanded:

- /Current View1

Name: Views

Splitter Ratio: 0.5

Visualization Manager:

Class: ""

Displays:

- Alpha: 0.5

Cell Size: 1

Class: rviz_default_plugins/Grid

Color: 160; 160; 164

Enabled: true

Line Style:

Line Width: 0.029999999329447746

Value: Lines

Name: Grid

Normal Cell Count: 0

Offset:

X: 0

Y: 0

Z: 0

Plane: XY

Plane Cell Count: 10

Reference Frame: <Fixed Frame>

Value: true

- Angle Tolerance: 0.10000000149011612

Class: rviz_default_plugins/Odometry

Covariance:

Orientation:

Alpha: 0.5

Color: 255; 255; 127

Color Style: Unique

Frame: Local

Offset: 1

Scale: 1

Value: true

Position:

Alpha: 0.30000001192092896

Color: 204; 51; 204

Scale: 1

Value: true

Value: true

Enabled: false

Keep: 100

Name: Odometry

Position Tolerance: 0.10000000149011612

Shape:

Alpha: 1

Axes Length: 1

Axes Radius: 0.10000000149011612

Color: 255; 25; 0

Head Length: 0.10000000149011612

Head Radius: 0.10000000149011612

Shaft Length: 0.10000000149011612

Shaft Radius: 0.05000000074505806

Value: Arrow

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /odom

Value: false

- Alpha: 0.699999988079071

Class: rviz_default_plugins/Map

Color Scheme: map

Draw Behind: false

Enabled: true

Name: Map

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /map

Update Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /map_updates

Use Timestamp: false

Value: true

- Alpha: 1

Autocompute Intensity Bounds: true

Autocompute Value Bounds:

Max Value: 10

Min Value: -10

Value: true

Axis: Z

Channel Name: intensity

Class: rviz_default_plugins/LaserScan

Color: 255; 255; 255

Color Transformer: Intensity

Decay Time: 0

Enabled: true

Invert Rainbow: false

Max Color: 255; 255; 255

Max Intensity: 0

Min Color: 0; 0; 0

Min Intensity: 0

Name: LaserScan

Position Transformer: XYZ

Selectable: true

Size (Pixels): 3

Size (m): 0.009999999776482582

Style: Flat Squares

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Best Effort

Value: /scan

Use Fixed Frame: true

Use rainbow: true

Value: true

- Class: rviz_default_plugins/TF

Enabled: true

Frame Timeout: 15

Frames:

All Enabled: true

base_link:

Value: true

camera_back:

Value: true

camera_front:

Value: true

imu_link:

Value: true

map:

Value: true

odom:

Value: true

tugbot/scan_back/scan_back:

Value: true

tugbot/scan_front/scan_front:

Value: true

tugbot/scan_omni/scan_omni:

Value: true

warnign_light:

Value: true

wheel_left:

Value: true

wheel_right:

Value: true

Marker Scale: 1

Name: TF

Show Arrows: true

Show Axes: true

Show Names: true

Tree:

map:

odom:

base_link:

camera_back:

{}

camera_front:

{}

imu_link:

{}

tugbot/scan_back/scan_back:

{}

tugbot/scan_front/scan_front:

{}

tugbot/scan_omni/scan_omni:

{}

warnign_light:

{}

wheel_left:

{}

wheel_right:

{}

Update Interval: 0

Value: true

- Class: rviz_default_plugins/MarkerArray

Enabled: true

Name: MarkerArray

Namespaces:

slam_toolbox: true

slam_toolbox_edges: true

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /slam_toolbox/graph_visualization

Value: true

- Alpha: 1

Class: rviz_default_plugins/RobotModel

Collision Enabled: false

Description File: ""

Description Source: Topic

Description Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /robot_description

Enabled: true

Links:

All Links Enabled: true

Expand Joint Details: false

Expand Link Details: false

Expand Tree: false

Link Tree Style: Links in Alphabetic Order

base_link:

Alpha: 1

Show Axes: false

Show Trail: false

Value: true

camera_back:

Alpha: 1

Show Axes: false

Show Trail: false

camera_front:

Alpha: 1

Show Axes: false

Show Trail: false

imu_link:

Alpha: 1

Show Axes: false

Show Trail: false

tugbot/scan_back/scan_back:

Alpha: 1

Show Axes: false

Show Trail: false

tugbot/scan_front/scan_front:

Alpha: 1

Show Axes: false

Show Trail: false

tugbot/scan_omni/scan_omni:

Alpha: 1

Show Axes: false

Show Trail: false

Value: true

warnign_light:

Alpha: 1

Show Axes: false

Show Trail: false

wheel_left:

Alpha: 1

Show Axes: false

Show Trail: false

Value: true

wheel_right:

Alpha: 1

Show Axes: false

Show Trail: false

Value: true

Mass Properties:

Inertia: false

Mass: false

Name: RobotModel

TF Prefix: ""

Update Interval: 0

Value: true

Visual Enabled: true

- Alpha: 1

Autocompute Intensity Bounds: true

Autocompute Value Bounds:

Max Value: 10

Min Value: -10

Value: true

Axis: Z

Channel Name: intensity

Class: rviz_default_plugins/PointCloud2

Color: 255; 255; 255

Color Transformer: Intensity

Decay Time: 0

Enabled: true

Invert Rainbow: false

Max Color: 255; 255; 255

Max Intensity: 0

Min Color: 0; 0; 0

Min Intensity: 0

Name: PointCloud2

Position Transformer: XYZ

Selectable: true

Size (Pixels): 3

Size (m): 0.009999999776482582

Style: Flat Squares

Topic:

Depth: 5

Durability Policy: Volatile

Filter size: 10

History Policy: Keep Last

Reliability Policy: Reliable

Value: /world/world_demo/model/tugbot/link/scan_omni/sensor/scan_omni/scan/points

Use Fixed Frame: true

Use rainbow: true

Value: true

Enabled: true

Global Options:

Background Color: 48; 48; 48

Fixed Frame: map

Frame Rate: 1

Name: root

Tools:

- Class: rviz_default_plugins/MoveCamera

- Class: rviz_default_plugins/Select

- Class: rviz_default_plugins/FocusCamera

- Class: rviz_default_plugins/Measure

Line color: 128; 128; 0

- Class: rviz_default_plugins/SetInitialPose

Covariance x: 0.25

Covariance y: 0.25

Covariance yaw: 0.06853891909122467

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /initialpose

- Class: rviz_default_plugins/SetGoal

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /move_base_simple/goal

- Class: rviz_default_plugins/PublishPoint

Single click: true

Topic:

Depth: 5

Durability Policy: Volatile

History Policy: Keep Last

Reliability Policy: Reliable

Value: /clicked_point

Transformation:

Current:

Class: rviz_default_plugins/TF

Value: true

Views:

Current:

Class: rviz_default_plugins/Orbit

Distance: 21.67657470703125

Enable Stereo Rendering:

Stereo Eye Separation: 0.05999999865889549

Stereo Focal Distance: 1

Swap Stereo Eyes: false

Value: false

Focal Point:

X: -0.1385374516248703

Y: 2.8173320293426514

Z: -0.014920835383236408

Focal Shape Fixed Size: true

Focal Shape Size: 0.05000000074505806

Invert Z Axis: false

Name: Current View

Near Clip Distance: 0.009999999776482582

Pitch: 0.7297971248626709

Target Frame: <Fixed Frame>

Value: Orbit (rviz)

Yaw: 0.9353978633880615

Saved: ~

Window Geometry:

Displays:

collapsed: false

Height: 1136

Hide Left Dock: false

Hide Right Dock: false

QMainWindow State:
000000ff00000000fd00000004000000000000015600000416fc020000000afb000000120053006500c00650063
00740069006f006e00000001e10000009b0000005c00fffffffb0000001e0054006f006f006c002000500072006f0070
006500720074006900650073020000001ed000001df00000185000000a3fb0000001200560069006500770073002
00054006f006f02000001df000002110000018500000122fb000000200054006f006f006c002000500072006f0070
006500720074006900650073003203000002880000011d000002210000017afb00000010004400690073007000
6c0061007900730100000003d00000416000000c900fffffffb0000002000730065006c0065006300740069006f006e
00200062007500660066006500720200000138000000aa0000023a00000294fb00000014005700690064006500
530074006500720065006f02000000e6000000d2000003ee0000030bfb0000000c004b0069006e0065006300740
200000186000001060000030c00000261fb0000000a0049006d00610067006500000000230000000e0000000000
0000000fb0000000a0049006d00610067006500000002cb000001380000000000000000000000010000010f0000
0416fc0200000003fb0000001e0054006f006f006c002000500072006f0070006500720074006900650073010000
00410000007800000000000000000fb0000000a00560069006500770073010000003d00000416000000a400fffffffb
000000120053006500c0065006300740069006f006e010000025a000000b20000000000000000000000020000
0490000000a9fc0100000001fb0000000a00560069006500770073030000004e00000080000002e100000197000
00003000004420000003efc0100000002fb0000000800540069006d006501000000000000044200000000000000
00fb0000000800540069006d00650100000000000004500000000000000000000004c900000416000000040000
00040000000800000008fc00000001000000020000000100000000a0054006f006f006c00730100000000ffffffff0000
000000000000

Selection:

collapsed: false

Tool Properties:

collapsed: false

Views:

collapsed: false

Width: 1850

X: 70

Y: 27),CMakeLists.txt(cmake_minimum_required(VERSION 3.8)

project(tugbot_slam)

if(CMAKE_COMPILER_IS_GNUCXX OR CMAKE_CXX_COMPILER_ID MATCHES "Clang")

add_compile_options(-Wall -Wextra -Wpedantic)

endif()

find dependencies

find_package(ament_cmake REQUIRED)

```
# uncomment the following section in order to fill in

# further dependencies manually.

# find_package(<dependency> REQUIRED)

if(BUILD_TESTING)

find_package(ament_lint_auto REQUIRED)

# the following line skips the linter which checks for copyrights

# comment the line when a copyright and license is added to all source files

set(ament_cmake_copyright_FOUND TRUE)

# the following line skips cpplint (only works in a git repo)

# comment the line when this package is in a git repo and when

# a copyright and license is added to all source files

set(ament_cmake_cpplint_FOUND TRUE)

ament_lint_auto_find_test_dependencies()

endif()

install(DIRECTORY

launch

config

rviz

DESTINATION share/${PROJECT_NAME}/

)

ament_package()),package.xml(<?xml version="1.0"?>

<?xml-model href="http://download.ros.org/schema/package_format3.xsd"
schematypens="http://www.w3.org/2001/XMLSchema"?>

<package format="3">

<name>tugbot_slam</name>

<version>0.0.0</version>

<description>TODO: Package description</description>

<maintainer email="porizou.electro@gmail.com">porizou</maintainer>

<license>TODO: License declaration</license>

<buildtool_depend>ament_cmake</buildtool_depend>
```

<test_depend>ament_lint_auto</test_depend>

<test_depend>ament_lint_common</test_depend>

<export>

<build_type>ament_cmake</build_type>

</export>

</package>)